Strong correlations, entanglement and quantum phases

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Chapter 1

Introduction

The central question that motivates this thesis is the following: under what conditions can a system of many interacting spins, arranged on a two-dimensional lattice, transition between qualitatively distinct phases of matter? Phase transitions are familiar in everyday experience — water boils, ferromagnets lose their magnetisation above the Curie point — but the question of *when* a phase transition is even *allowed* to occur is far from obvious. Why does the Ising model on a one-dimensional chain possess no ordered phase at any non-zero temperature, while the same model on a two-dimensional lattice does? The answer, due to Peierls [1], depends on the geometry of the underlying lattice through a competition between energy and entropy. The argument also makes clear what is meant by symmetry breaking, an order parameter, and a critical point, and it provides the conceptual bridge to the *quantum* phase transition that occupies the rest of this thesis.

1.1 Ground states and excitations of the classical Ising model

Consider the classical Ising model on a d-dimensional hypercubic lattice with N sites, defined by

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \qquad \sigma_i = \pm 1, \qquad J > 0. \tag{1.1}$$

At zero temperature there are two degenerate ground states, $|\uparrow\uparrow \cdots \uparrow\rangle$ and $|\downarrow\downarrow \cdots \downarrow\rangle$, both with energy

$$E_0 = -NdJ, \tag{1.2}$$

since each of the N sites contributes d bonds (the lattice coordination number is $2d$, and the bond count Nd avoids double-counting). The lowest excitation flips a single spin; in d dimensions this spin has $2d$ neighbours and each of the $2d$ bonds it shares now costs $+J$

instead of $-J$, giving

$$E_1 = E_0 + 2d \cdot 2J = E_0 + 4dJ. \tag{1.3}$$

Excitations therefore raise the energy, and at strictly zero temperature only the two ground states are occupied. The question is what happens at finite temperature, and whether the macroscopic distinction between the two ground states survives thermal fluctuations.

1.2 Peierls' droplet argument

The cleanest way to answer this is to ask: starting from a fully ordered ferromagnetic background, what is the free energy of a "droplet" of the minority phase? If droplets of arbitrarily large size lower the free energy, they proliferate and destroy the order; if not, the ordered phase is stable.

One dimension. A droplet of L consecutive flipped spins in a one-dimensional chain has exactly two domain walls — one at each end of the flipped region. Each domain wall corresponds to one broken ferromagnetic bond contributing $2J$, so the energy of the droplet is

$$\Delta E_{\mathrm{1D}} = 4J, \tag{1.4}$$

independent of the droplet size L. The entropy gain comes from the freedom to place the droplet anywhere along a chain of L sites,

$$\Delta S_{\mathrm{1D}} = k_B \ln L, \tag{1.5}$$

and the free energy is

$$F_{\mathrm{1D}} \;=\; \Delta E - T\Delta S \;=\; 4J - Tk_B \ln L. \tag{1.6}$$

For any $T > 0$, $F_{\mathrm{1D}} \to -\infty$ as $L \to \infty$: the system always lowers its free energy by admitting arbitrarily large droplets. Droplets proliferate, ferromagnetic order is destroyed, and the one-dimensional Ising model exhibits **no finite-temperature phase transition**.

Two dimensions. A droplet of the minority phase is now bounded by a *closed loop* of length L on the dual lattice, and each link of this perimeter corresponds to a broken bond contributing $2J$. The energy cost is

$$\Delta E_{\mathrm{2D}} = 2JL, \tag{1.7}$$

linear in the perimeter. Counting the number of admissible droplet boundaries requires more care. A naive estimate is that at each step of the boundary walk there are four

nearest-neighbour directions (north, south, east, west) available, giving 4^L walks of length L. Several physical constraints reduce this number:

- *Closure.* After L steps the walk must return to its starting point, which is a non-trivial global constraint.
- *No backtracking.* The walk cannot reverse direction on consecutive steps, eliminating one of the four directions at each step.
- *No self-intersection.* The walk must be self-avoiding (no link is traversed twice).

After incorporating these constraints, the number of admissible closed self-avoiding walks of length L grows asymptotically as c^L, where the *connectivity constant* c of the square lattice is bounded by

$$2 < c < 3. \tag{1.8}$$

The "no backtracking" rule alone reduces the bound from 4 to 3, and self-avoidance reduces it further to a value above 2 but below 3. The exact value of c is not analytically known, but the bound is enough for the argument. The entropy of a single droplet is therefore

$$\Delta S_{\mathrm{2D}} \approx k_B L \ln c, \tag{1.9}$$

and the free energy becomes

$$F_{\mathrm{2D}} \;=\; \Delta E - T\Delta S \;=\; L\Big(2J - Tk_B \ln c\Big), \tag{1.10}$$

which is *linear* in L. The sign of the bracketed coefficient determines the fate of the ordered phase:

- If $Tk_B \ln c \;<\; 2J$ (low temperature), F_{2D} grows with L and large droplets are exponentially suppressed. Ferromagnetic order is stable.
- If $Tk_B \ln c > 2J$ (high temperature), F_{2D} becomes arbitrarily negative as $L \to \infty$. Droplets of all sizes proliferate and order is destroyed.

The crossover defines a critical temperature

$$k_B T_c \;\sim\; \frac{2J}{\ln c}. \tag{1.11}$$

Using the bounds $2 < c < 3$ yields the quantitative prediction

$$\frac{2J}{\ln 3} \;<\; k_B T_c \;<\; \frac{2J}{\ln 2}, \qquad \text{i.e.} \quad 1.82 \;<\; \frac{k_B T_c}{J} \;<\; 2.89. \tag{1.12}$$

That this estimate captures the right physics can be checked against the exact result.

Kramers and Wannier [2] located the critical point using a duality argument, and Onsager [3] gave the closed-form solution of the 2D Ising model. The exact critical temperature is

$$\frac{k_B T_c}{J} \;=\; \frac{2}{\ln\!\left(1+\sqrt{2}\right)} \;\approx\; 2.269, \tag{1.13}$$

which lies comfortably inside the Peierls bounds. The Peierls argument is therefore not only qualitative but predictive to within a small factor. Lattice dimensionality is not a cosmetic detail: it changes the scaling of the boundary energy with droplet size from constant ($d = 1$) to linear ($d = 2$), and with it the qualitative possibility of long-range order at non-zero temperature.

1.3 Symmetry, order parameter, and spontaneous symmetry breaking

The two ground states $|\uparrow\uparrow\cdots\rangle$ and $|\downarrow\downarrow\cdots\rangle$ are related by the global spin-flip operation $\sigma_j \to -\sigma_j$ for every site j. For the quantum problem we shall study, this operation is implemented by the unitary

$$\mathcal{U} \;=\; \prod_{j=1}^{N} \sigma_j^x, \tag{1.14}$$

which is its own inverse, $\mathcal{U}^2 = \mathbb{1}$, and therefore Hermitian. Its action on the Pauli operators is

$$\mathcal{U}^\dagger \sigma_j^z \mathcal{U} = -\sigma_j^z, \qquad \mathcal{U}^\dagger \sigma_j^y \mathcal{U} = -\sigma_j^y, \qquad \mathcal{U}^\dagger \sigma_j^x \mathcal{U} = +\sigma_j^x. \tag{1.15}$$

Because each term in the Ising Hamiltonian is bilinear in σ^z (and any quantum transverse-field term is linear in σ^x), the Hamiltonian is invariant:

$$\mathcal{U}^\dagger H \mathcal{U} = H, \qquad [\,\mathcal{U},\, H\,] = 0. \tag{1.16}$$

The set $\{\mathbb{1}, \mathcal{U}\}$ forms a $\mathbb{Z}_2$ symmetry group of the model. In an ordered ferromagnetic ground state with $\langle \sigma_j^z \rangle \neq 0$, the symmetry $\mathcal{U}$ is *spontaneously broken*: the ground state itself is not invariant under $\mathcal{U}$ even though the Hamiltonian is.

The technical definition of spontaneous symmetry breaking proceeds via a careful order of limits. Add to the Hamiltonian a small symmetry-breaking field $h_\parallel$ that explicitly favours one of the two ground states,

$$H \;\to\; H - h_\parallel \sum_{i=1}^{N} \sigma_i^z. \tag{1.17}$$

Compute $\langle \sigma_i^z \rangle$ as a function of $h_\parallel$ and the system size N. The symmetry is said to be

spontaneously broken if

$$\lim_{h_\parallel \to 0^+} \lim_{N \to \infty} \langle \sigma_i^z \rangle \neq 0. \tag{1.18}$$

The order of limits is essential: taking $N \to \infty$ first allows the symmetry-broken sector to be picked out, after which $h_\parallel$ can be sent to zero without the system tunnelling between the two ground states. Reversing the order yields zero for any finite system. The expectation value $\langle \sigma_i^z \rangle$ is the **order parameter** of the transition: it is non-zero in the ferromagnetic phase and zero in the disordered phase. Other models with continuous symmetries — the XY model with U(1) and the Heisenberg model with SU(2) — admit analogous order parameters and are characterised by their own symmetry-breaking patterns and universality classes.

1.4 From classical thermal transitions to quantum phase transitions

The Peierls argument as stated above concerns a classical model driven by thermal fluctuations. The model studied in this thesis is the *quantum* transverse-field Ising model (TFIM), whose Hamiltonian is

$$H = -\sum_{\langle i,j \rangle} J_{ij}\, \sigma_i^z \sigma_j^z - h \sum_i \sigma_i^x, \tag{1.19}$$

and whose phase transition occurs at zero temperature, driven by the transverse field h rather than by temperature. The two stories are nevertheless linked through the **path-integral formulation of quantum statistical mechanics**.

Via a Suzuki–Trotter decomposition [4] of the imaginary-time evolution operator $e^{-\beta H}$, the partition function of a d-dimensional quantum spin model at inverse temperature β can be rewritten exactly as that of a $(d+1)$-dimensional classical model. The extra dimension corresponds to imaginary time, and its length is set by β. Taking the zero-temperature limit $\beta \to \infty$ extends this additional dimension infinitely, so that a d-dimensional quantum system at $T = 0$ is mapped onto a $(d+1)$-dimensional classical system at finite effective temperature.

For the TFIM the mapping is particularly transparent. The two-dimensional quantum TFIM at $T = 0$ corresponds to a three-dimensional anisotropic classical Ising model at finite temperature. The Peierls argument applied to that three-dimensional classical model — droplet surfaces of area $\sim L^2$ contributing energy $\sim JL^2$, against an entropy of the same order — again guarantees a finite-temperature transition. Pulled back to the original quantum problem, this transition becomes the quantum phase transition of the

2D TFIM, tuned by the ratio h/J. The same reasoning applied to the *one-dimensional* quantum TFIM at $T = 0$ produces a 2D classical Ising model at finite T, whose Onsager solution again guarantees a transition. Both 1D and 2D quantum TFIMs therefore exhibit quantum phase transitions, but they belong to different universality classes: the 1D model is in the $(1 + 1)$D Ising class (exactly solvable, central charge $c = 1/2$), while the 2D quantum model belongs to the $(2 + 1)$D Ising universality class with critical exponents $\beta \approx 0.326$ and $\nu \approx 0.630$ [25].

1.5 Existence of the quantum phase transition

The phase diagram of the previous section asserts that, at zero temperature, the two-dimensional transverse-field Ising model undergoes a quantum phase transition at a finite critical field $(h/J)_c$. It is worth pausing on a question that is easy to ask and surprisingly subtle to answer: how does one *know* that a genuine phase transition occurs at all, rather than a smooth crossover in which the magnetisation merely decreases rapidly, but continuously, from its ordered value to zero? A steep slope in $\langle \sigma^z \rangle$ is not by itself evidence of a transition: an analytic function can fall from one to nearly zero as sharply as one likes and still possess no singular point whatsoever. This section establishes, with the level of rigour appropriate to an introduction, why a transition is nonetheless guaranteed.

1.5.1 Non-analyticity as the defining criterion

The precise criterion for a phase transition is the *non-analyticity* of the relevant thermodynamic potential as a function of the control parameter. At finite temperature this potential is the free-energy density; at zero temperature its role is played by the ground-state energy density

$$e_0(h) \;=\; \lim_{N\to\infty} \frac{E_0(h)}{N}, \tag{1.20}$$

and a quantum phase transition at h_c is, by definition, a point at which $e_0(h)$ fails to be an analytic function of h.

Two observations make this definition non-trivial. First, for any *finite* system the ground-state energy $E_0(h)$ is the lowest eigenvalue of a finite Hermitian matrix whose entries depend smoothly on h; as long as the ground state remains separated from the rest of the spectrum by a gap, analytic perturbation theory (the Kato–Rellich theorem) guarantees that $E_0(h)$ is analytic in h. A finite system therefore exhibits no transition: the singularity can emerge only in the thermodynamic limit $N \to \infty$, where it arises from the closing of the spectral gap.

Second, it is essential that the criterion is the loss of *analyticity*, not merely of smoothness. An infinitely differentiable function may vanish on an entire interval and yet be non-zero

elsewhere – the standard example being $f(x) = e^{-1/x}$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$, which is smooth but does not equal its (identically zero) Taylor series at the origin. An *analytic* function cannot behave in this way. This is the content of the *identity theorem*: if a function analytic on a connected interval vanishes on any sub-interval, it vanishes identically on the whole interval. The proof is short. If f and all its derivatives vanish at a point, its Taylor series is identically zero and, by analyticity, f itself vanishes in a neighbourhood of that point; the set of points at which f and all its derivatives vanish is therefore simultaneously open and closed, and by connectedness it must be the entire interval.

The relevance is immediate. The longitudinal magnetisation, defined through the order of limits

$$m \;=\; \lim_{h_\parallel \to 0^+} \; \lim_{N \to \infty} \langle \sigma_i^z \rangle, \tag{1.21}$$

is *exactly* zero throughout the disordered phase and *non-zero* throughout the ordered phase. No analytic function can take the value zero on a whole interval and a non-zero value on another; the identity theorem forbids it. Consequently, if the model genuinely possesses two such phases, then $m(h)$ – and with it $e_0(h)$ – cannot be analytic over the entire range of h, and at least one non-analytic point must exist. It remains only to establish that the two qualitatively distinct phases are real.

1.5.2 The two limiting phases from perturbation theory

Both limits of the model are controlled, and each is reached by a convergent perturbative expansion built upon a gapped ground state.

Small field ($h \ll J$): the ordered phase. Writing $H = H_0 + H'$ with $H_0 = -J \sum_{\langle ij \rangle} \sigma_i^z \sigma_j^z$ and $H' = -h \sum_i \sigma_i^x$, the unperturbed ground state is the ferromagnet $|\psi_0\rangle = |\uparrow\uparrow \cdots \uparrow\rangle$ with energy $E_0 = -NdJ$, the up branch being selected by an infinitesimal longitudinal field. Because σ_i^x flips a single spin, $H'|\psi_0\rangle$ couples $|\psi_0\rangle$ only to the single-flip states $|i\rangle \equiv |\uparrow \cdots \downarrow_i \cdots \uparrow\rangle$, each of which lies an energy $4dJ$ above the ground state: the flipped spin breaks its $2d$ bonds, and each broken bond changes from $-J$ to $+J$ at a cost of $2J$. First-order perturbation theory then gives the corrected state

$$|\psi\rangle \;=\; |\uparrow\uparrow \cdots \uparrow\rangle \;+\; \frac{h}{4dJ} \sum_i |i\rangle \;+\; \mathcal{O}(h^2), \tag{1.22}$$

from which the longitudinal magnetisation follows, after computing both the numerator and the norm to second order, as

$$\langle \sigma_i^z \rangle \;=\; \frac{\langle \psi | \sigma_i^z | \psi \rangle}{\langle \psi | \psi \rangle} \;=\; 1 - 2\left(\frac{h}{4dJ}\right)^2 + \mathcal{O}(h^4). \tag{1.23}$$

The magnetisation is reduced from its saturated value but remains strictly positive: the small-field phase is ordered, with $m \neq 0$. Crucially, because the unperturbed ground state is gapped – the gap being $4dJ$ to the single-flip excitations – the Kato–Rellich theorem guarantees that this expansion converges in a disk of finite radius about $h = 0$. The ordered phase therefore occupies a finite interval $0 \leq h < h_1$, throughout which $e_0(h)$ is analytic; the expansion is valid not at the single point $h = 0$ but on an entire neighbourhood of it.

Large field ($h \gg J$): the disordered phase. In the opposite limit the field term dominates, $H_0 = -h \sum_i \sigma_i^x$, and the unperturbed ground state is the product of spins polarised along the field,

$$|\psi_0\rangle \;=\; \bigotimes_i \tfrac{1}{\sqrt{2}}(|\uparrow\rangle_i + |\downarrow\rangle_i)\,, \tag{1.24}$$

which is *unique* and separated from all excited states by a gap of order $2h$. Here the vanishing of the order parameter is not merely perturbative but exact, and it follows from symmetry. The Hamiltonian commutes with the global spin-flip $U = \prod_i \sigma_i^x$, which satisfies $U\sigma_i^z U = -\sigma_i^z$. A unique ground state must be an eigenstate of U, so $U|\psi\rangle = \pm|\psi\rangle$, and therefore

$$\langle\sigma_i^z\rangle = \langle\psi|\, U\sigma_i^z\, U\,|\psi\rangle = -\langle\sigma_i^z\rangle \quad\Longrightarrow\quad \langle\sigma_i^z\rangle = 0. \tag{1.25}$$

This argument applies at *every* field strength at which the ground state is unique, and is not an order-by-order statement. Since the gap keeps the ground state unique over a finite interval $h > h_2$, the order parameter is exactly zero throughout that interval: the large-field phase is disordered, with $m = 0$.

1.5.3 The transition is forced

The two analyses establish more than two isolated limits. A non-zero spectral gap is precisely what allows a phase to occupy a finite interval rather than a single point: a gapped ground state evolves analytically with h, so each expansion converges on a neighbourhood and the phase is a genuine, extended region. The two phases are, however, qualitatively incompatible. In the disordered phase the ground state is unique and the order parameter vanishes by symmetry, equation (1.25); in the ordered phase the order parameter is non-zero, equation (1.23), which in the thermodynamic limit necessarily requires a degenerate ground space, i.e. spontaneously broken symmetry. One cannot pass continuously from a unique, gapped ground state with $m = 0$ to a degenerate one with $m \neq 0$ while keeping the gap open. The gap must therefore close at some finite h_c lying between the two phases.

The same conclusion follows directly from the identity theorem. The order parameter $m(h)$ is exactly zero on the interval $h > h_2$ and strictly positive on the interval $0 \leq h < h_1$. Were

$m(h)$ analytic on the whole range $0 < h < \infty$, its vanishing on a sub-interval would force it to vanish everywhere, contradicting $m \neq 0$ at small field. Hence $m(h)$ is non-analytic at some $h_c \in [h_1, h_2]$, and since $m = -\partial e_0/\partial h_\parallel$, the ground-state energy density $e_0(h)$ is itself non-analytic there. By the criterion of Section 1.5.1, this non-analyticity *is* the quantum phase transition, and its existence is guaranteed. Equivalently, the closing of the gap is the single event that terminates the radius of convergence of *both* perturbative expansions, renders $e_0(h)$ non-analytic, and constitutes the transition.

Two remarks complete the argument. First, perturbation theory establishes the *form* of each phase order by order; the genuinely non-perturbative input is that the ordered phase persists over a finite range of field – equivalently, that the small-field series has a finite, rather than vanishing, radius of convergence. This is supplied by the Peierls droplet argument of Section 1.2, applied through the quantum-to-classical mapping to the three-dimensional classical Ising model, whose ordered phase is rigorously established. Second, the argument guarantees the *existence* of h_c but not its value. In one dimension the model is exactly solvable: the excitation gap closes linearly as $\Delta = 2|h - J|$ at $h_c = J$, the second derivative of $e_0(h)$ diverges logarithmically, and the order parameter vanishes as $m = \left(1 - (h/J)^2\right)^{1/8}$ for $h < J$ and is exactly zero beyond, furnishing an explicit realisation of the non-analyticity [25]. In two dimensions no closed form exists, and the critical field is fixed numerically by quantum Monte Carlo at $(h/J)_c = 3.04438(2)$ [6], the value used as a benchmark throughout this thesis.

1.6 Quantum–Classical Correspondence

The Peierls argument of Section 1.2 and the brief remark in Section 1.4 asserted that the transverse-field Ising model in d dimensions is tied, through imaginary time, to a classical Ising model in $d + 1$ dimensions. This section makes that statement precise. The result is the foundation for the one-dimensional study of later chapters: it is exactly what guarantees that the $T = 0$ quantum chain possesses a genuine phase transition, since the $d = 1$ image is the two-dimensional classical Ising model solved by Onsager [3].

Write the Hamiltonian (1.19) as a sum of two non-commuting pieces,

$$\hat{H} \;=\; \hat{H}_0 + \hat{H}_1, \qquad \hat{H}_0 = -J\sum_{\langle i,j\rangle} \sigma_i^z \sigma_j^z, \qquad \hat{H}_1 = -h\sum_i \sigma_i^x, \tag{1.26}$$

the Ising coupling $\hat{H}_0$ and the transverse field $\hat{H}_1$. The "problem" with the quantum partition function $Z = \operatorname{Tr} e^{-\beta\hat{H}}$ is precisely that $[\hat{H}_0, \hat{H}_1] \neq 0$, so the Boltzmann operator does not factorise. The Suzuki–Trotter construction [4] trades this non-commutativity for one extra classical dimension.

Trotter decomposition. Fix the inverse temperature $\beta = 1/k_B T$, choose a large integer M, and set

$$\varepsilon = \frac{\beta}{M}, \qquad \varepsilon M = \beta, \tag{1.27}$$

so that $Z = \mathrm{Tr}\,(e^{-\varepsilon \hat{H}})^M$. For $\varepsilon \ll 1$ the first-order Trotter splitting

$$e^{-\varepsilon \hat{H}} \;=\; e^{-\varepsilon \hat{H}_0}\, e^{-\varepsilon \hat{H}_1} \;+\; \mathcal{O}(\varepsilon^2) \tag{1.28}$$

holds, and the error vanishes as $M \to \infty$ at fixed β. Hence

$$Z \;=\; \mathrm{Tr}\Big(e^{-\varepsilon \hat{H}_0}\, e^{-\varepsilon \hat{H}_1}\Big)^M \;+\; \mathcal{O}(\varepsilon). \tag{1.29}$$

Imaginary-time slices. Evaluate the trace in the σ^z product basis, inserting a resolution of the identity

$$\mathbb{1} \;=\; \sum_{\{\sigma_i = \pm 1\}} |\{\sigma_i\}\rangle\langle\{\sigma_i\}| \tag{1.30}$$

between each of the M factors. Labelling the slices by $\ell = 0, 1, \ldots, M-1$ and assigning to each an imaginary-time coordinate $\tau_\ell = \ell\varepsilon \in \{0, \varepsilon, 2\varepsilon, \ldots, \beta - \varepsilon\}$, the cyclicity of the trace imposes the periodic seam $\sigma_i(\tau_M) \equiv \sigma_i(\tau_0)$. The partition function becomes a product of transfer matrix elements between neighbouring slices,

$$Z \;=\; \left[\prod_{\ell=0}^{M-1} \sum_{\{\sigma_i(\tau_\ell)\}}\right] \prod_{\ell=0}^{M-1} \Big\langle \{\sigma_i(\tau_\ell + \varepsilon)\}\Big|\, e^{-\varepsilon \hat{H}_0}\, e^{-\varepsilon \hat{H}_1}\, \Big|\{\sigma_i(\tau_\ell)\}\Big\rangle. \tag{1.31}$$

The index τ is a genuinely new coordinate — an "imaginary-time" or "expansion" direction of total extent β — so that the d-dimensional quantum problem has acquired a $(d+1)$-th dimension.

The single-slice matrix element. Consider one factor of (1.31). Because the states $|\{\sigma_i\}\rangle$ are σ^z eigenstates, the Ising factor $e^{-\varepsilon \hat{H}_0} = e^{\varepsilon J \sum_{\langle ij\rangle} \sigma_i^z \sigma_j^z}$ is *diagonal* and contributes the classical Boltzmann weight $\exp\!\Big(\varepsilon J \sum_{\langle ij\rangle} \sigma_i \sigma_j\Big)$, i.e. a spatial coupling $K_1 = \varepsilon J$ within the slice. The transverse factor factorises over sites,

$$\Big\langle \{\sigma_i'\}\Big| e^{\varepsilon h \sum_i \sigma_i^x}\Big|\{\sigma_i\}\Big\rangle = \prod_{i=1}^{N} \Big\langle \sigma_i'\Big| e^{\varepsilon h \sigma_i^x}\Big|\sigma_i\Big\rangle \equiv \prod_{i=1}^{N} C_{\sigma_i' \sigma_i}. \tag{1.32}$$

Since $(\sigma^x)^2 = \mathbb{1}$, the exponential collapses to $e^{\varepsilon h \sigma^x} = \cosh(\varepsilon h)\,\mathbb{1} + \sinh(\varepsilon h)\,\sigma^x$, and as σ^x flips the σ^z eigenvalue,

$$C_{\sigma'\sigma} = \cosh(\varepsilon h)\,\delta_{\sigma'\sigma} + \sinh(\varepsilon h)\,(1 - \delta_{\sigma'\sigma}) = \begin{cases} \cosh(\varepsilon h), & \sigma' = \sigma, \\ \sinh(\varepsilon h), & \sigma' \neq \sigma. \end{cases} \tag{1.33}$$

Writing this single-bond weight in Ising form $C_{\sigma'\sigma} = A\, e^{K_2\,\sigma'\sigma}$ with $\sigma', \sigma = \pm 1$ and matching the two cases,

$$A\, e^{K_2} = \cosh(\varepsilon h), \qquad A\, e^{-K_2} = \sinh(\varepsilon h) \implies A = \sqrt{\tfrac{1}{2}\sinh(2\varepsilon h)}, \qquad K_2 = \tfrac{1}{2}\ln\coth(\varepsilon h). \tag{1.34}$$

Collecting the diagonal Ising weight and the per-site transverse weight, the generic slice matrix element is therefore

$$\left\langle \{\sigma_i'\} \middle| e^{-\varepsilon \hat{H}_0} e^{-\varepsilon \hat{H}_1} \middle| \{\sigma_i\} \right\rangle = A^N \exp\left[K_2 \sum_{i=1}^{N} \sigma_i'\,\sigma_i \;+\; K_1 \sum_{\langle ij \rangle} \sigma_i\,\sigma_j \right]. \tag{1.35}$$

The K_2 term links the *same* spatial site i on adjacent slices τ and $\tau + \varepsilon$ (an imaginary-time bond); the K_1 term links spatial nearest neighbours $\langle ij \rangle$ at *equal* τ.

The $(d+1)$-dimensional classical Ising model. Substituting (1.35) into (1.31) and gathering all M slices yields

$$\boxed{Z = A^{MN} \sum_{\{\sigma_i(\tau_\ell)\}} \exp\left[K_2 \sum_{\ell=0}^{M-1} \sum_{i=1}^{N} \sigma_i(\tau_\ell)\,\sigma_i(\tau_\ell + \varepsilon) \;+\; K_1 \sum_{\ell=0}^{M-1} \sum_{\langle ij \rangle} \sigma_i(\tau_\ell)\,\sigma_j(\tau_\ell) \right]} \tag{1.36}$$

with $K_1 = \varepsilon J$, $K_2 = \tfrac{1}{2}\ln\coth(\varepsilon h)$ and $A = \sqrt{\tfrac{1}{2}\sinh(2\varepsilon h)}$. Equation (1.36) is *exactly* the partition function of a classical Ising model on a $(d+1)$-dimensional lattice: the d original spatial directions, with coupling K_1, together with one imaginary-time direction of extent β and periodic boundary conditions, with coupling K_2. Each classical spin $\sigma_i(\tau_\ell) = \pm 1$ lives at a site i of the spatial lattice and a slice τ_ℓ of imaginary time (Figure 1.1). The quantum fluctuations driven by the transverse field h have been converted into ferromagnetic correlations along the new time direction.

For the one-dimensional quantum chain ($d = 1$) that is the focus of this thesis, the image is a *two*-dimensional anisotropic classical Ising model. This is precisely the model solved exactly by Onsager [3]: its finite-temperature order–disorder transition, pulled back through the correspondence, is the zero-temperature quantum phase transition of the transverse-field chain at $h/J = 1$.

Anisotropy and universality. The two couplings are markedly different. As $\varepsilon \to 0$,

$$K_1 = \varepsilon J \to 0, \qquad K_2 = \tfrac{1}{2}\ln\coth(\varepsilon h) \sim -\tfrac{1}{2}\ln(\varepsilon h) \to \infty, \tag{1.37}$$

so the resulting classical model is highly anisotropic — weakly coupled in space and strongly coupled in imaginary time. This anisotropy is, however, an *irrelevant* perturbation at the Ising critical point: it rescales the spatial and temporal directions differently but does

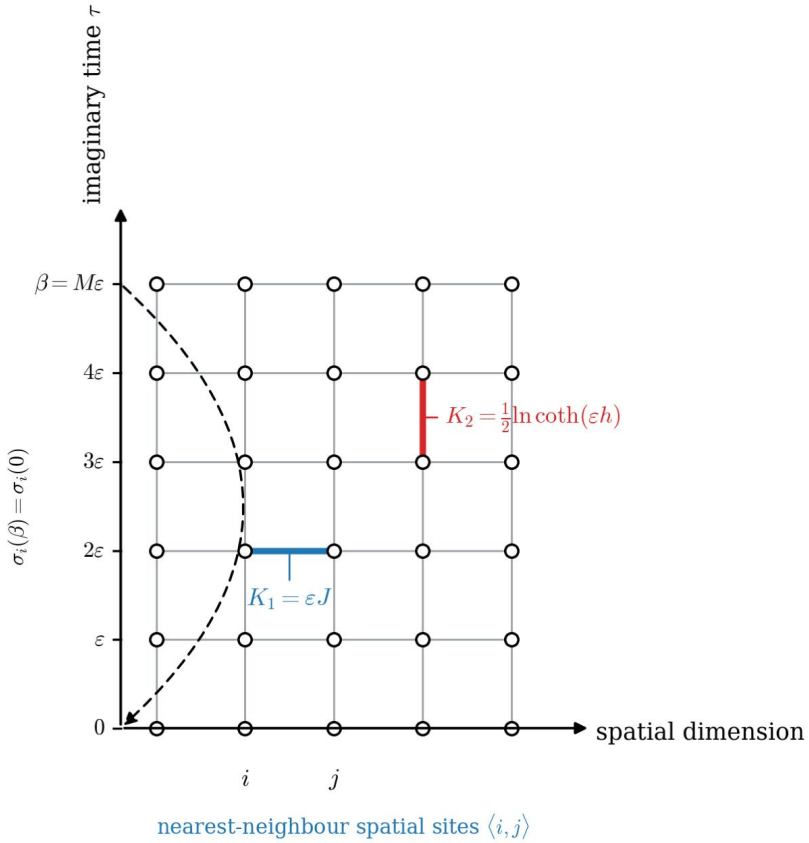


Figure 1.1: The classical lattice underlying Eq. (1.36), here drawn for a one-dimensional quantum chain ($d = 1$). The horizontal axis runs over the spatial sites of the chain, with nearest neighbours $\langle i, j\rangle$; the vertical axis is the imaginary-time direction $\tau = 0, \varepsilon, 2\varepsilon, \ldots, \beta = M\varepsilon$. Equal-time spatial bonds (blue) carry the coupling $K_1 = \varepsilon J$, while imaginary-time bonds (red), which connect the *same* spatial site on two adjacent slices, carry $K_2 = \frac{1}{2}\ln\coth(\varepsilon h)$. The cyclicity of the trace makes the time direction periodic, $\sigma_i(\beta) = \sigma_i(0)$ (dashed seam). The $(1 + 1)$-dimensional grid shown here is precisely the anisotropic classical Ising lattice solved exactly by Onsager.

not change the universality class. The anisotropic classical model therefore flows to the same critical theory as the isotropic $(d+1)$-dimensional Ising model, and the quantum and classical transitions share critical exponents — the $(1+1)$D Ising values (central charge $c = \frac{1}{2}$) for the chain, and the $(2+1)$D Ising values $\beta \approx 0.326$, $\nu \approx 0.630$ quoted in Section 1.4 for the square lattice.

More generally, the construction above shows that the quantum statistical mechanics of any d-dimensional model can be reformulated as the classical statistical mechanics of a model in $d+1$ dimensions, the extra dimension being imaginary time of extent $\beta = 1/T$. The zero-temperature quantum limit $\beta \to \infty$ extends this direction to infinity, so that a d-dimensional quantum system at $T = 0$ maps onto a $(d+1)$-dimensional classical system at a finite effective temperature. This is the precise sense in which the quantum critical point of the transverse-field Ising model inherits the physics of a classical Ising transition one dimension higher.

1.7 Plan of the thesis

The two-dimensional transverse-field Ising model is therefore the simplest setting in which dimensionality permits a quantum phase transition driven by a non-commuting field. The remainder of this thesis is concerned with computing, as accurately as is practical, the ground-state energy, longitudinal and transverse magnetisations, and entanglement entropy of this model on both sides of the transition and through the critical region. We use tensor-network states — specifically infinite projected entangled-pair states (iPEPS) — as the variational ansatz, and exploit GPU acceleration to reach bond dimensions large enough that the physics, rather than the algorithm, controls the accuracy of the results.

Chapter 2

Motivation

Condensed matter physics studies the collective behaviour of large numbers of interacting quantum particles. The properties one wishes to compute in any concrete model — and the ones we focus on in this thesis — are a small set of equilibrium observables:

- the **ground-state energy per site**, which fixes the energy scale of the phase and underpins every other equilibrium quantity;
- the **magnetisation**, both the longitudinal component $\langle \sigma^z \rangle$ that serves as the order parameter and the transverse component $\langle \sigma^x \rangle$ that responds to the applied field;
- the **entanglement entropy**, a measure of how much quantum information is shared between a region and its complement, which diagnoses critical behaviour even when conventional order parameters fail to distinguish phases.

The obstacle to computing any of these directly is the size of the quantum Hilbert space. A lattice of N spin-$\frac{1}{2}$ sites has 2^N basis states, growing exponentially with system size. Exact diagonalisation is therefore restricted to small clusters (around 30 sites in two dimensions), and the thermodynamic limit must be approached through approximate methods — perturbation theory, mean-field treatments, quantum Monte Carlo, and tensor-network states. Each of these makes a different compromise between accuracy and scalability.

2.1 The transverse-field Ising model

The simplest interacting quantum many-body model that already shows non-trivial behaviour is the **transverse-field Ising model** (TFIM). On a square lattice it is defined

by the Hamiltonian (1.19), reproduced here for convenience,

$$H \;=\; -\sum_{\langle i,j\rangle} J_{ij}\,\sigma_i^z\sigma_j^z \;-\; h\sum_i \sigma_i^x, \tag{2.1}$$

where σ_i^x and σ_i^z are Pauli operators on site i, $J_{ij} > 0$ is a ferromagnetic Ising exchange (taken nearest-neighbour and uniform in this work), and h is a uniform transverse field. The two terms do not commute: the Ising term, written in the z-eigenbasis, favours classical spin alignment, while the field term flips spins between the two z-eigenstates. This non-commutation is the source of all quantum behaviour in the model.

2.2 Three reasons it deserves study

(i) It exhibits a quantum phase transition. As h/J is increased from small to large values at $T = 0$, the ground state changes from a ferromagnet with broken $\mathbb{Z}_2$ symmetry and finite longitudinal magnetisation $\langle\sigma^z\rangle \neq 0$ to a paramagnetic product state $|+,+,\ldots\rangle$ in which $\langle\sigma^z\rangle = 0$. The transition is driven purely by quantum fluctuations — no thermal energy is supplied — and on the square lattice occurs at a critical ratio $(h/J)_c \approx 3.044$ [6]. The TFIM is therefore one of the cleanest theoretical settings in which to study a quantum phase transition, free of disorder, frustration or thermal smearing.

(ii) It is the entry point to the path-integral formulation of quantum statistical mechanics. Via a Suzuki–Trotter decomposition of the partition function, a d-dimensional quantum system at zero temperature can be rewritten as a $(d+1)$-dimensional classical statistical mechanics problem, with the extra dimension corresponding to imaginary time. For the TFIM the mapping is particularly transparent: the d-dimensional quantum TFIM maps onto a $(d+1)$-dimensional anisotropic classical Ising model. This quantum-to-classical correspondence is the basis of path-integral and quantum Monte Carlo techniques, and the TFIM is the textbook example for understanding how the mapping works.

(iii) It is experimentally realised. The Hamiltonian above accurately describes the magnetic insulator $\mathbf{LiHoF_4}$, in which Holmium ions form an Ising ferromagnet whose lowest-energy doublet acts as an effective spin-$\frac{1}{2}$. A magnetic field applied perpendicular to the easy axis plays the role of the transverse field, and the resulting quantum phase transition has been observed and characterised experimentally [5].

These three properties — a clean quantum phase transition, a direct link to classical statistical mechanics, and a concrete experimental realisation — make the TFIM the natural first model on which to test any new numerical method for two-dimensional

quantum lattice systems. In this thesis we use it both as an object of physical study and as the benchmark against which our GPU-accelerated tensor-network implementation is validated.

Chapter 3

The Density Matrix Renormalisation Group

The Density Matrix Renormalisation Group, in its modern formulation, is a variational algorithm over the space of matrix product states (MPS) of fixed bond dimension D. The starting observation is that the full many-body Hilbert space of N spins on a chain has dimension d^N, which grows exponentially with system size and is intractable beyond roughly 20 sites by exact diagonalisation. An MPS of bond dimension D instead has only $\mathcal{O}(NdD^2)$ parameters and forms a manifold of states within the full Hilbert space whose dimension scales *linearly* with N. The variational task is therefore to find, within this manifold, the state of lowest energy.

3.1 Variational space of matrix product states

We assume the MPS is brought into **site-canonical form** with orthogonality centre at site ℓ. In this gauge the state can be written as

$$|\Psi\rangle \;=\; \sum_{\alpha,\sigma_\ell,\beta} \Big[C_\ell\Big]^{\alpha\sigma_\ell\beta}\; |\alpha\rangle \otimes |\sigma_\ell\rangle \otimes |\beta\rangle\,, \tag{3.1}$$

where $|\alpha\rangle$ and $|\beta\rangle$ are the orthonormal left and right Schmidt bases generated by the left-normalised A-tensors for sites $1,\dots,\ell-1$ and the right-normalised B-tensors for sites $\ell+1,\dots,N$, and σ_ℓ is the physical index at the centre site. The composite Schmidt indices α and β each run over D values, and σ_ℓ over the local physical dimension d. All variational freedom in this gauge is concentrated in the single tensor C_ℓ, which carries D^2d complex amplitudes.

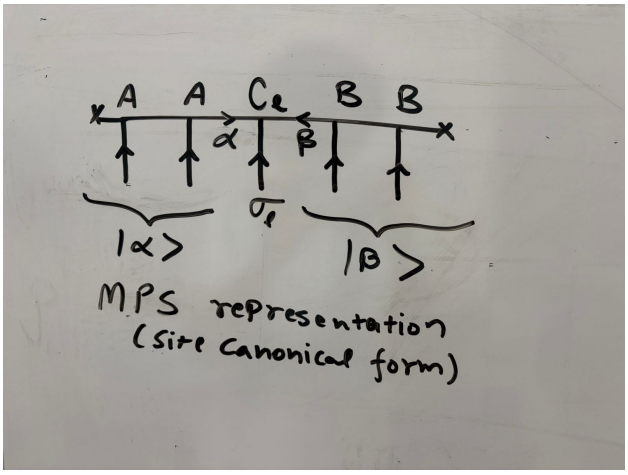


Figure 3.1: MPS in site-canonical form with orthogonality centre at site ℓ. Left-normalised A-tensors (left of C_ℓ) and right-normalised B-tensors (right of C_ℓ) carry the Schmidt bases $|\alpha\rangle$ and $|\beta\rangle$; all variational freedom is concentrated in the single centre tensor C_ℓ, which carries the physical index σ_ℓ and virtual indices α, β each of dimension D.

The Hamiltonian is encoded as a **matrix product operator** (MPO):

$$\hat{H} \;=\; \sum_{\vec{\sigma},\vec{\sigma}\,'} |\vec{\sigma}\,'\rangle \left(\prod_{\ell} \big[W_\ell\big]^{\sigma'_\ell}{}_{\sigma_\ell} \right) \langle\vec{\sigma}| \,. \tag{3.2}$$

Each W_ℓ is a matrix of operators carrying two physical indices (an outgoing σ'_ℓ upper and an incoming σ_ℓ lower) and two virtual indices μ_ℓ, ν_ℓ that are contracted between neighbouring MPO tensors. The same arrow convention is adopted on the MPO virtual bonds as on the MPS virtual bonds, so that the orientation of the MPS triangles is consistent throughout and most arrows can henceforth be dropped without ambiguity.

3.2 Lagrange multiplier formulation

The ground state is obtained by minimising $\langle\Psi|\,\hat{H}\,|\Psi\rangle$ over the variational manifold, subject to the unit-normalisation constraint

$$\langle\Psi|\Psi\rangle \;=\; 1. \tag{3.3}$$

Introducing a Lagrange multiplier λ, this is equivalent to extremising

$$\mathcal{L}[\Psi] \;=\; \langle\Psi|\,\hat{H}\,|\Psi\rangle \;-\; \lambda\,\langle\Psi|\Psi\rangle\,, \tag{3.4}$$

where λ will turn out to be the ground-state energy. The strategy of DMRG is to perform

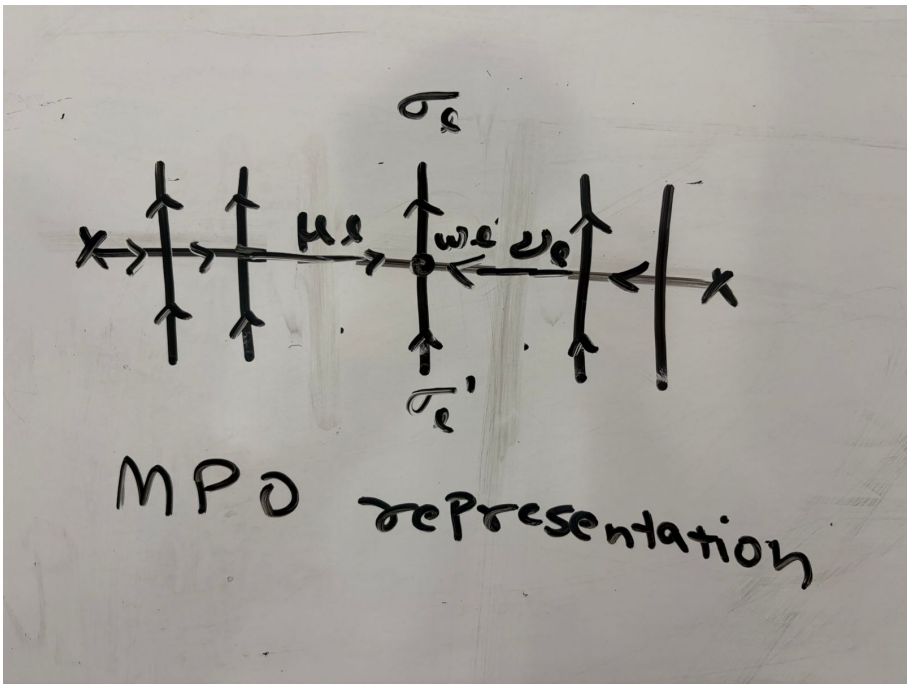


Figure 3.2: Matrix product operator (MPO) representation of the Hamiltonian $\hat{H}$. Each site tensor W_ℓ carries an outgoing physical index σ'_ℓ (top), an incoming physical index σ_ℓ (bottom), and two virtual bond indices μ_ℓ, ν_ℓ (left and right) that are contracted between neighbouring MPO tensors along the chain.

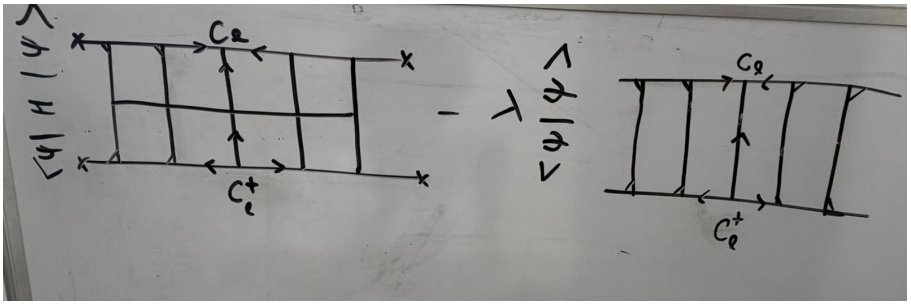


Figure 3.3: Tensor-network diagram for the variational functional $\mathcal{L}[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle - \lambda\langle\Psi|\Psi\rangle$. The first term (left network) contracts the bra MPS, the MPO, and the ket MPS layer by layer; the second term (right network) subtracts λ times the MPS norm. Setting the derivative with respect to $C_\ell^\dagger$ to zero yields the effective single-site eigenvalue problem at the orthogonality centre.

this extremisation **one tensor at a time** — holding every A_k for $k < \ell$ and every B_k for $k > \ell$ fixed, and varying only C_ℓ. The stationary condition is

$$\frac{\partial}{\partial C_\ell^\dagger}\Big[\,\langle\Psi|\,\hat{H}\,|\Psi\rangle - \lambda\,\langle\Psi|\Psi\rangle\,\Big] \;=\; 0. \tag{3.5}$$

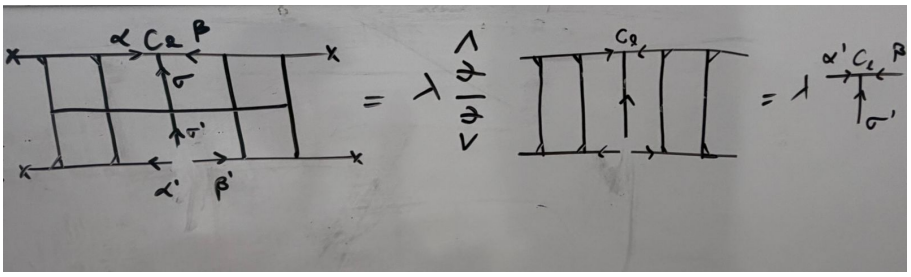


Figure 3.4: Tensor-network form of the stationary condition $\partial\mathcal{L}/\partial C_\ell^\dagger = 0$. Differentiating removes $C_\ell^\dagger$ from the bra layer, leaving an open slot. The left-hand network (action of $\hat{H}_\ell^{(1)}$ on C_ℓ) is set equal to λ times the right-hand network (norm overlap), expressing the eigenvalue equation for the optimum centre tensor.

3.3 Effective eigenvalue problem

Taking the derivative with respect to $C_\ell^\dagger$ removes the bra-side $C_\ell^\dagger$ from the tensor network and leaves an open slot of the same shape as C_ℓ. Closing the resulting "zippers" from the left and right — i.e. contracting the network of $A^\dagger \cdot W \cdot A$ environments to the left of site ℓ and of $B^\dagger \cdot W \cdot B$ environments to the right — produces an effective single-site equation,

$$\hat{H}_\ell^{(1)}\, C_\ell \;=\; \lambda\, C_\ell, \tag{3.6}$$

together with the normalisation

$$C_\ell^\dagger\, C_\ell \;=\; 1. \tag{3.7}$$

Here $\hat{H}_\ell^{(1)}$ is the **effective Hamiltonian** at site ℓ, obtained by contracting all tensors of the full network $\langle\Psi|\,\hat{H}\,|\Psi\rangle$ except the central C_ℓ and $C_\ell^\dagger$.

Reorganising indices makes the structure transparent. Treating C_ℓ as a vector labelled by the composite index

$$a \;=\; (\alpha,\, \sigma_\ell,\, \beta), \tag{3.8}$$

and $\hat{H}_\ell^{(1)}$ as a matrix with one composite row index a' and one composite column index a, equation (3.6) becomes

$$\Big[\hat{H}_\ell^{(1)}\Big]^{a'}{}_{a}\,\Big[C_\ell\Big]^{a} \;=\; \lambda\,\Big[C_\ell\Big]^{a'}, \qquad \Big[C_\ell^\dagger\Big]_{a}\,\Big[C_\ell\Big]^{a} \;=\; 1, \tag{3.9}$$

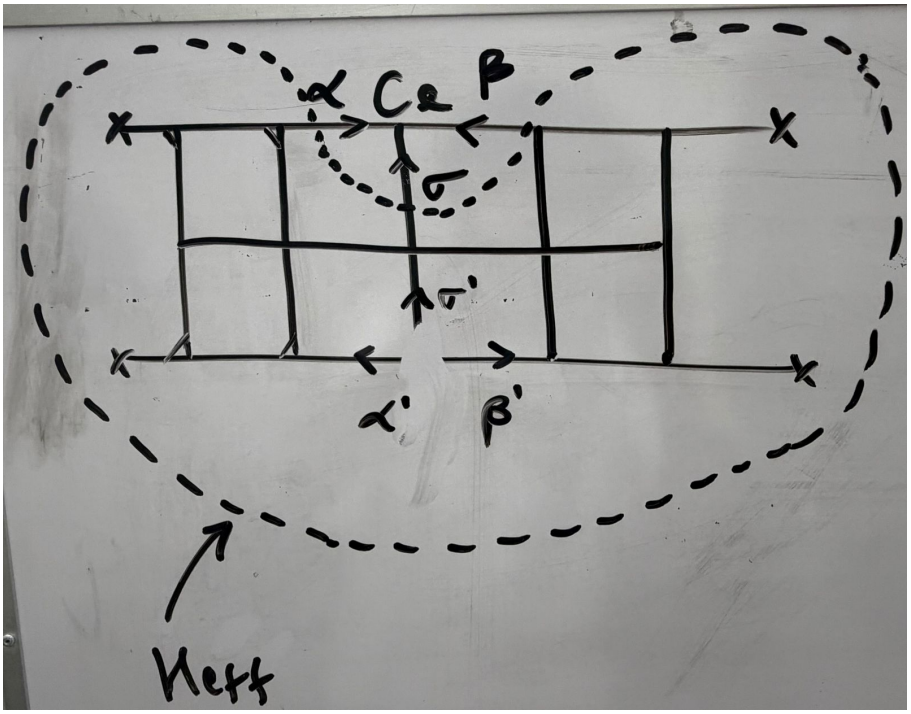


Figure 3.5: Effective Hamiltonian $\hat{H}_\ell^{(1)}$ (shaded boundary) acting on the centre tensor C_ℓ. All tensors of the full network $\langle\Psi|\hat{H}|\Psi\rangle$ except C_ℓ and $C_\ell^\dagger$ are contracted into a single effective operator of dimension $(D^2 d) \times (D^2 d)$, whose lowest eigenvector is C_ℓ and whose lowest eigenvalue is the variational ground-state energy.

which is a standard finite-dimensional Hermitian eigenvalue problem (compare the analogous expression in the MPS reference [11]). The dimension of the effective Hamiltonian is

$$\dim\!\left(\hat{H}_\ell^{(1)}\right) \;=\; \left(D^2\,d\right) \times \left(D^2\,d\right), \tag{3.10}$$

i.e. polynomially large in D rather than exponentially large in N. The lowest eigenvector is computed not by full diagonalisation but by an iterative Krylov-subspace method such as the **Lanczos algorithm**, which requires only repeated application of $\hat{H}_\ell^{(1)}$ to a trial vector and converges to the ground state in a small number of iterations.

If the MPS is *not* in site-canonical form, the same derivation yields a generalised eigenvalue problem

$$\hat{H}_\ell^{(1)}\, C_\ell \;=\; \lambda\, G_\ell^{(1)}\, C_\ell, \tag{3.11}$$

where $G_\ell^{(1)}$ is the corresponding norm-matrix obtained from the right-hand side of (3.5) under the same derivative. Site-canonical form is preferred precisely because it reduces $G_\ell^{(1)}$ to the identity and recovers (3.6).

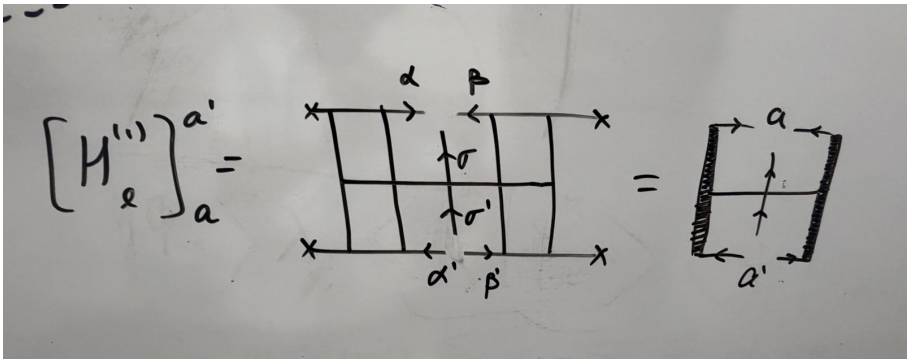


Figure 3.6: Component form of the effective single-site eigenvalue equation. The composite index $a = (\alpha, \sigma_\ell, \beta)$ fuses the two virtual and one physical leg of C_ℓ into a single vector label. The matrix $[\hat{H}^{(1)}_\ell]^{a'}_{a}$ is obtained by contracting the left environment block, the MPO tensor W_ℓ, and the right environment block, reducing the full infinite network to a standard finite-dimensional Hermitian eigenvalue problem.

3.4 The Lanczos method for the local eigenvalue problem

The effective single-site eigenvalue equation (3.6) is Hermitian and accessed only through repeated applications of $\hat{H}^{(1)}_\ell$ to a vector — exactly the setting in which the **Lanczos algorithm** [7, 8, 9] is optimal. We outline only the parts of the algorithm that matter for understanding why DMRG converges as fast as it does.

The minimisation of $E[|\psi\rangle] = \langle\psi|\,\hat{H}\,|\psi\rangle \,/\, \langle\psi|\psi\rangle$ along the steepest-descent direction generated by the functional gradient $\delta E/\delta\,\langle\psi|$ leads, after L iterations starting from an arbitrary vector $|v_0\rangle$, to a search restricted to the **Krylov space**

$$K_L(|v_0\rangle) \;=\; \mathrm{span}\Big\{|v_0\rangle\,,\; \hat{H}\,|v_0\rangle\,,\; \hat{H}^2\,|v_0\rangle\,,\; \ldots,\; \hat{H}^L\,|v_0\rangle\Big\}, \tag{3.12}$$

of dimension $L+1$. Orthogonalising the iterates by Gram–Schmidt produces an orthonormal Krylov basis $\{|v_0\rangle\,, |v_1\rangle\,, \ldots, |v_L\rangle\}$ satisfying the **three-term recursion**

$$\hat{H}\,|v_n\rangle \;=\; |v_{n-1}\rangle\; b_n \;+\; |v_n\rangle\; a_n \;+\; |v_{n+1}\rangle\; b_{n+1}, \tag{3.13}$$

with coefficients

$$a_n \;=\; \langle v_n|\,\hat{H}\,|v_n\rangle\,, \qquad b_n \;=\; \langle v_n|\,\hat{H}\,|v_{n-1}\rangle\,. \tag{3.14}$$

The recursion has only three terms because $\hat{H}\,|v_j\rangle \in \mathrm{span}\{|v_{j-1}\rangle\,, |v_j\rangle\,, |v_{j+1}\rangle\}$, so $\langle v_n|\,\hat{H}\,|v_j\rangle = 0$ whenever $|n-j|>1$. As a result only three vectors need to be held in memory at any time.

In this basis the Hamiltonian acquires a real symmetric tridiagonal form

$$H_{K_L} \;=\; \begin{pmatrix} a_0 & b_1 & & & \\ b_1 & a_1 & b_2 & & \\ & b_2 & a_2 & \ddots & \\ & & \ddots & \ddots & b_L \\ & & & b_L & a_L \end{pmatrix}, \tag{3.15}$$

which is diagonalised in $\mathcal{O}(L^2)$ operations by standard tridiagonal solvers. Its lowest eigenpair $(E_g^L,\, \boldsymbol{\psi}_g^L)$ yields the best approximation to the true ground-state energy and ground-state vector within K_L,

$$\left|\psi_g^L\right\rangle \;=\; \sum_{j=0}^{L} |v_j\rangle \; (\psi_g^L)^j, \tag{3.16}$$

which can be assembled on the fly by restarting the Lanczos iteration from $|v_0\rangle$ once the coefficients $(\psi_g^L)^j$ are known.

The quality of the approximation is monitored by the **residual energy variance**

$$\tau[|\psi\rangle] \;:=\; \left\| (\hat{H} - E)\,|\psi\rangle \right\|^2 \;=\; \langle\psi|\,\hat{H}^2\,|\psi\rangle - \langle\psi|\,\hat{H}\,|\psi\rangle^2\,, \tag{3.17}$$

and the iteration is terminated when τ drops below a chosen threshold. The convergence of E_g^L to the exact E_g is exponential in L, controlled by the spectral gap Δ to the first excited state:

$$\left|E_g^L - E_g\right| \;\sim\; e^{-c\,L\,\sqrt{\Delta}}. \tag{3.18}$$

This is the second reason — beyond the entanglement area law of the next section — that DMRG performs so well on gapped one-dimensional systems. A modest bond dimension D is sufficient to *represent* the ground state, *and* the local eigensolver converges in only $\mathcal{O}(1/\sqrt{\Delta})$ Lanczos iterations to reach a fixed accuracy. In gapless or higher-dimensional systems both costs deteriorate simultaneously: D must grow to accommodate the additional entanglement, and Lanczos convergence slows as $\Delta \to 0$.

3.5 The sweep algorithm

Solving (3.9) at site ℓ yields the optimal centre tensor at that site, call it C_ℓ^G, together with the current variational estimate $\lambda \simeq E_0$ of the ground-state energy. To proceed to the next site, the orthogonality centre is shifted from ℓ to $\ell + 1$ by absorbing C_ℓ^G into the

right-neighbour tensor $B_{\ell+1}$ and performing a singular value decomposition,

$$C_\ell^G \, B_{\ell+1} \;=\; U \, S \, V^\dagger \, B_{\ell+1} \;=\; \tilde{A}_\ell \, C_{\ell+1}, \tag{3.19}$$

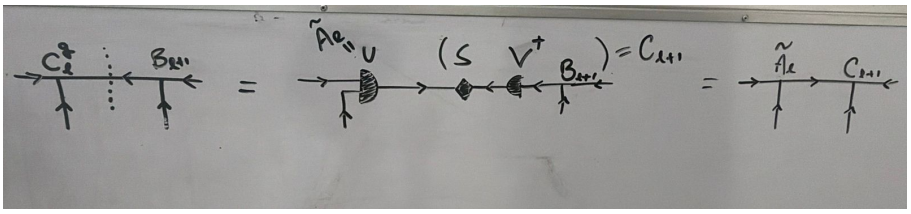


Figure 3.7: DMRG sweep step: shifting the orthogonality centre from site ℓ to $\ell+1$. The optimised centre tensor C_ℓ^G is absorbed into the right-neighbour tensor $B_{\ell+1}$ and singular-value-decomposed as $C_\ell^G B_{\ell+1} = USV^\dagger B_{\ell+1}$. The factor $\tilde{A}_\ell = U$ is left-normalised and absorbed into the chain; the new centre tensor $C_{\ell+1} = SV^\dagger B_{\ell+1}$ is then optimised at the next step.

where $\tilde{A}_\ell = U$ is now left-normalised and $C_{\ell+1} = S\,V^\dagger\,B_{\ell+1}$ is the new centre tensor at site $\ell+1$. One then optimises $C_{\ell+1}$ by repeating (3.9) with the freshly contracted environment, then $C_{\ell+2}$, and so on.

A **sweep** consists of carrying this local optimisation from one end of the chain to the other; the algorithm then sweeps back and forth until the ground-state energy stops changing within a chosen tolerance. In practice, one observes the error $E_g - E_g^{\text{exact}}$ decay rapidly for the first few sweeps at a given bond dimension D, plateau at a D-dependent residual of order 10^{-6}, then drop again whenever D is increased. Iterating in this way until both the energy and the bond-dimension extrapolation have converged yields the variational ground state to controlled accuracy.

3.6 Why DMRG works so well in one dimension: the area law

The reason DMRG is essentially the method of choice for one-dimensional gapped quantum systems traces back to the structure of entanglement in their ground states. For a contiguous bipartition of a 1D chain into a region A of length L_A and its complement $\bar{A}$, the von Neumann entropy of the reduced density matrix $\rho_A = \mathrm{tr}_{\bar{A}} \, |\Psi\rangle\langle\Psi|$ obeys an **area law**:

$$S(\rho_A) \;=\; -\,\mathrm{tr}\big[\rho_A \ln \rho_A\big] \;\le\; S_{\max}, \tag{3.20}$$

where $S_{\max}$ is a constant independent of L_A. In one dimension the "area" of the boundary between A and $\bar{A}$ consists of just two points (or one for a half-infinite chain), so the

entropy is bounded by a constant rather than growing with the size of the region. Hastings proved this rigorously for gapped local Hamiltonians [10]: $S(\rho_A) \leq c\,\xi$, where ξ is the correlation length and c a constant depending on the spectral gap.

An MPS of bond dimension D can carry at most $S = \ln D$ units of entanglement across any bond, so to represent a state obeying the 1D area law faithfully it is sufficient to take D *independent of system size*. This is why DMRG converges in N rather than blowing up: the resource cost is controlled by the entanglement of the target state, not by the dimension of the Hilbert space.

For *critical* one-dimensional systems the area law is logarithmically violated, $S \sim (c/6)\ln L_A$ with c the central charge of the underlying conformal field theory, and a faithful MPS representation requires D to grow polynomially with system size — still tractable, although sub-optimal. The genuine difficulty arises in two dimensions, where even gapped ground states obey an area law in which the boundary of a region of linear size L scales as L itself, so that $S \sim L$ and the required bond dimension grows as

$$D \;\sim\; e^{L}. \tag{3.21}$$

This is the fundamental reason that DMRG, while spectacularly successful for one-dimensional chains and ladders with short-ranged interactions, becomes exponentially expensive in true two-dimensional geometries — motivating the projected entangled-pair state (PEPS) generalisation introduced later in this thesis.

3.7 Numerical Results: the Transverse-Field Ising Chain

We now apply the DMRG machinery of this chapter to the one-dimensional transverse-field Ising model as a controlled test case. The chain is an ideal benchmark for two reasons. First, it is exactly solvable by a Jordan–Wigner mapping to free fermions, so every DMRG number can be checked against an exact result rather than another approximation. Second, it possesses a continuous quantum phase transition at $h/J = 1$ in the (1+1)-dimensional Ising universality class, the $T = 0$ image of the two-dimensional classical Ising model of Section 1.6. The guiding observation of this section is that on a chain of *finite* length L there is no true transition — all observables are smooth, analytic functions of h — and the singular behaviour that *defines* the transition emerges only in the thermodynamic limit $L \to \infty$. The DMRG data make that statement quantitative.

Setup. We study the chain

$$\hat{H} \;=\; -J\sum_i \sigma_i^z\sigma_{i+1}^z \;-\; h\sum_i \sigma_i^x, \qquad J=1, \tag{3.22}$$

with open boundaries, using the matrix-product-state DMRG of this chapter: the bond dimension is ramped (as in the sweep algorithm of Section 3.5) until both the energy and the bond-dimension extrapolation are converged. Because the chain obeys a one-dimensional area law (Section 3.6), a modest bond dimension already saturates machine precision away from the critical point; only in the narrow critical window does the entanglement grow and a somewhat larger D become necessary, but it remains small throughout. We run chains of length $L = 16, 32, 64, 128$ over a grid of fields $h/J \in [0.2, 2.0]$ and measure, at each point, the ground-state energy per site, the order parameter, and the half-chain entanglement entropy. Every energy is compared against the exact open-chain free-fermion value.

3.7.1 Ground-state energy and benchmarking

The DMRG variationally minimises $\langle\hat{H}\rangle$ over the bond-D manifold. Figure 3.8 compares the converged energy per site with the exact free-fermion solution, and Figure 3.9 shows the corresponding error. The agreement is at the level of $|E_{\mathrm{DMRG}} - E_{\mathrm{exact}}| \lesssim 2.6\times10^{-7}$ across the entire field range, confirming that the variational state is genuinely converged and that the modest bond dimensions permitted by the area law are sufficient. The energy itself is a smooth, monotonically decreasing function of h with no visible feature at h_c: the transition is hidden in the analytic structure of $E_0(h)$, not in its value, and the diagnostics below are needed to expose it.

3.7.2 The order parameter and the finite-size crossover

The order parameter is the longitudinal magnetization $m = \langle\sigma^z\rangle$, which is non-zero in the ordered ($h < h_c$) phase and zero in the disordered ($h > h_c$) phase. On a finite chain with open boundaries the $\mathbb{Z}_2$ symmetry is not spontaneously broken, so $\langle\sigma^z\rangle = 0$ identically; we therefore extract m from the saturating value of the long-distance correlator, $m = \sqrt{\langle\sigma_i^z\sigma_j^z\rangle}$ at large $|i-j|$, which requires no symmetry-breaking field.

Figure 3.10 shows the result. At each finite L the order parameter is a *smooth, analytic* crossover from $m \approx 1$ deep in the ordered phase to $m \approx 0$ in the disordered phase. As L increases the crossover steepens about h_c and converges to the exact thermodynamic result [26]

$$m(h) \;=\; \begin{cases} \left(1-(h/J)^2\right)^{1/8}, & h < J, \\ 0, & h > J, \end{cases} \tag{3.23}$$

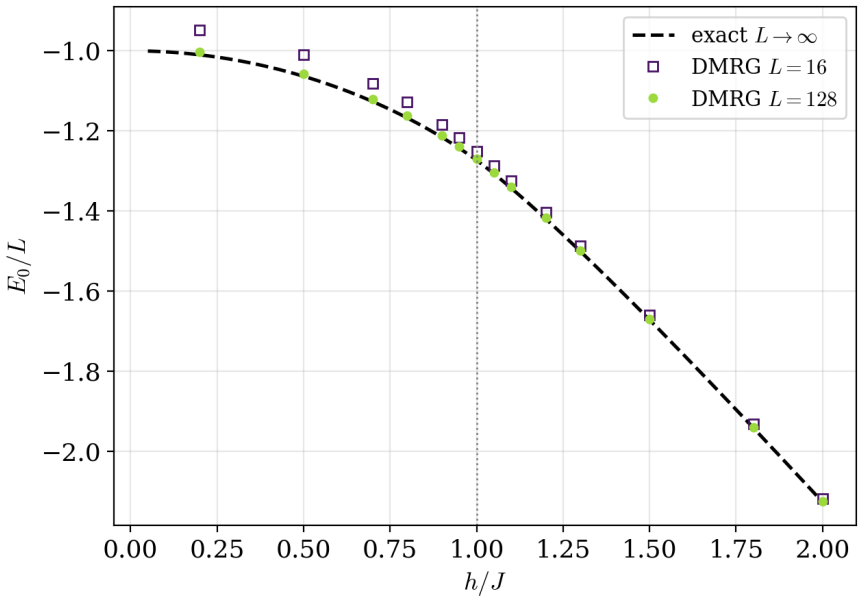


Figure 3.8: Ground-state energy per site of the TFIM chain: DMRG data (markers) against the exact thermodynamic free-fermion curve (dashed). The four chain lengths are indistinguishable at this resolution; the finite-L correction is an $O(1/L)$ boundary effect.

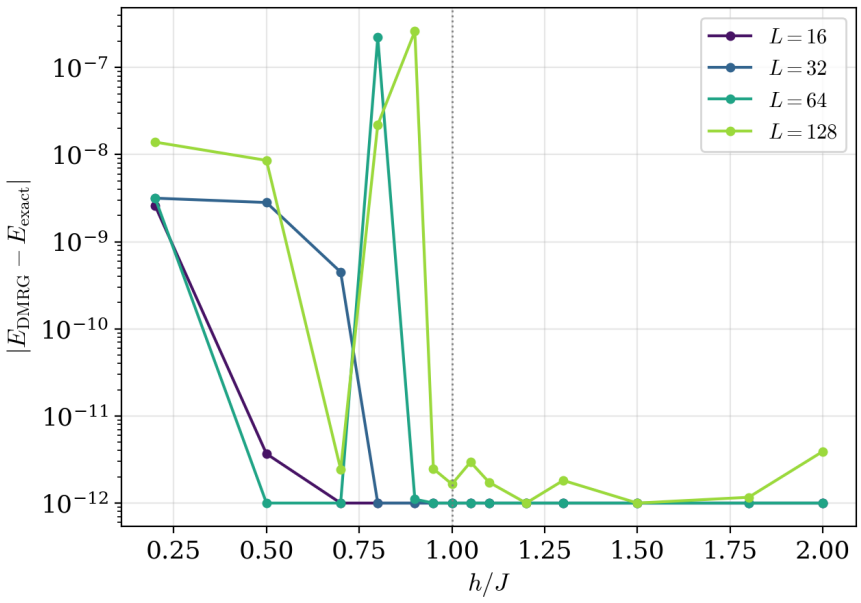


Figure 3.9: DMRG error relative to the exact open-chain free-fermion energy, for $L = 16, 32, 64, 128$. The error stays $\lesssim 10^{-7}$ across the field range, confirming convergence of the matrix-product-state ansatz.

a step at h_c whose vanishing is governed by the order-parameter exponent $\beta = \frac{1}{8}$ of the (1+1)D Ising class. The smoothness of $m(h)$ at every finite L *is* the statement that a finite system has no phase transition; the transition is precisely the $L \to \infty$ limit of this family of crossovers.

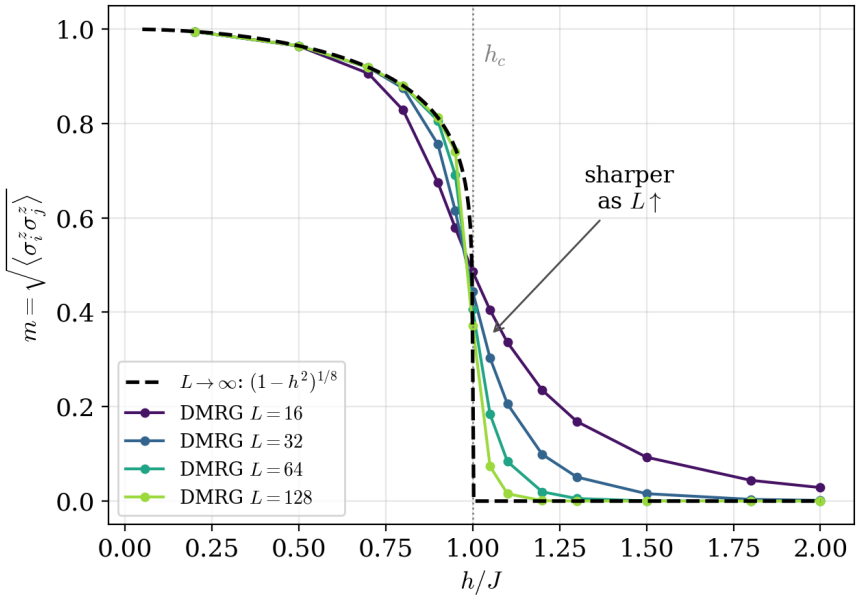


Figure 3.10: Order parameter $m = \sqrt{\langle \sigma_i^z \sigma_j^z \rangle}$ versus field (colour from dark to light with increasing L). The finite-L curves are smooth crossovers that steepen about h_c as L grows, converging to the exact step $(1-(h/J)^2)^{1/8}$ (dashed). The vanishing exponent is $\beta = \frac{1}{8}$.

3.7.3 Non-analyticity of the ground-state energy

At zero temperature the ground-state energy density $e_0(h) = E_0(h)/L$ plays the role of the free energy, and a quantum phase transition is, by definition, a point at which $e_0(h)$ fails to be analytic. Figure 3.11 plots the second derivative $-\,d^2e_0/dh^2$, the cleanest probe of that non-analyticity. Three facts explain why.

The first derivative is smooth. By the Hellmann–Feynman theorem the first derivative of the ground-state energy is an expectation value,

$$\frac{de_0}{dh} \;=\; \frac{1}{L}\left\langle \frac{\partial \hat{H}}{\partial h} \right\rangle \;=\; -\,\langle \sigma^x \rangle, \tag{3.24}$$

the transverse magnetization. This is finite and continuous through h_c: the transition is continuous (second order), so it leaves no signature in the first derivative.

The second derivative is a susceptibility. Differentiating once more, second-order perturbation theory in the field (with $\hat{V} = \sum_i \sigma_i^x$) gives

$$-\frac{d^2 e_0}{dh^2} \;=\; \frac{2}{L}\sum_{n\neq 0}\frac{\left|\langle n|\hat{V}|0\rangle\right|^2}{E_n - E_0} \;\geq\; 0. \tag{3.25}$$

This is the static susceptibility of the transverse magnetization — equivalently the structure factor of σ^x fluctuations in the ground state. Because every term is non-negative, $E_0(h)$ is concave in h and the curvature $-d^2e_0/dh^2$ is positive; this is why it, rather than $+d^2e_0/dh^2$, is the natural quantity to plot.

Why it is singular at h_c. The sum in (3.25) is dominated by the low-lying excitations, whose energy denominators $E_n - E_0$ are smallest. At the critical point the spectral gap $\Delta = E_1 - E_0$ closes (it scales as $\sim 1/L$ at finite L and vanishes as $L \to \infty$, Figure 3.12) while the correlation length diverges, so an ever-increasing number of states with small denominators and large matrix elements contribute. The susceptibility therefore diverges in the thermodynamic limit. At any finite L the gap stays open, the sum is finite, and the response is only a smooth, rounded peak — exactly the behaviour seen in Figure 3.11. The genuine singularity appears solely as $L \to \infty$.

The nature of the divergence follows from the quantum–classical correspondence of Section 1.6. There the $T = 0$ chain was shown to map onto the two-dimensional classical Ising model, under which $-d^2e_0/dh^2$ is the analogue of the classical *specific heat* $C \sim \partial^2 f/\partial T^2$. Onsager's exact solution [3] gives a *logarithmic* divergence of that specific heat (critical exponent $\alpha = 0$), so

$$-\frac{d^2 e_0}{dh^2} \;\sim\; -\ln|h - h_c| \qquad (L \to \infty), \tag{3.26}$$

a mild but genuine singularity. The finite-L curves in Figure 3.11 are the rounded precursors of this logarithmic spike, with $L = 512$ already indistinguishable from the thermodynamic limit. (The curve is computed from the exact free-fermion energy, which the DMRG reproduces to $\sim 10^{-7}$, so it is equally the DMRG prediction.)

3.7.4 Entanglement entropy and the central charge

The half-chain entanglement entropy $S = -\operatorname{Tr}(\rho_A \ln \rho_A)$, with ρ_A the reduced density matrix of one half of the chain, provides a third, entirely independent window on the transition — and clarifies what is meant by the "critical entropy".

Away from the critical point the gapped ground state obeys the one-dimensional area law of Section 3.6: S saturates to a constant independent of L. This is precisely why a fixed-D

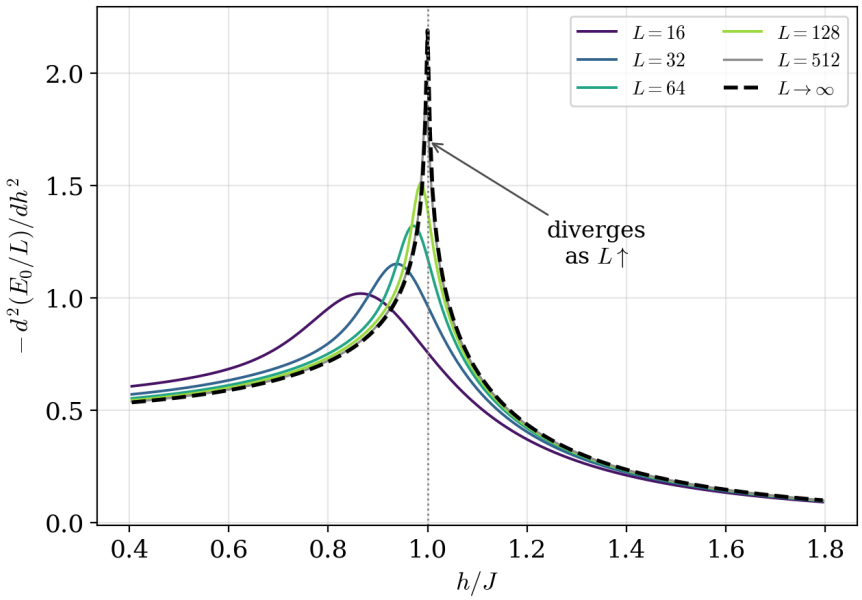


Figure 3.11: Curvature of the energy, $-d^2(E_0/L)/dh^2$, the analogue of the classical specific heat. Smooth rounded peaks at finite L sharpen into the logarithmic singularity at h_c only as $L \to \infty$ ($L = 512$ already coincides with the limit).

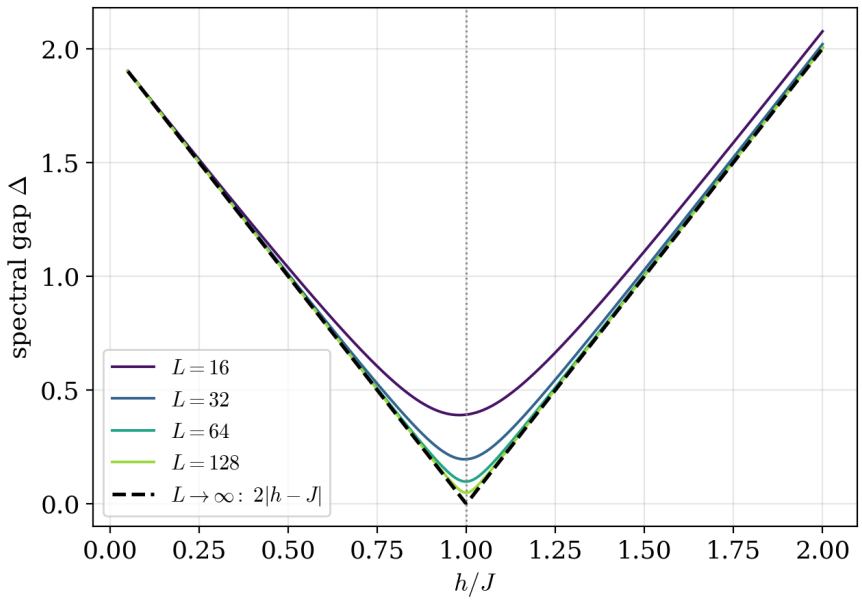


Figure 3.12: The spectral gap Δ, viewed from the finite-size momentum grid. It remains open at every finite L and closes only as $L \to \infty$, where it develops the non-analytic kink $2|h - J|$ that touches zero at h_c — the spectral counterpart of the energy singularity.

matrix product state represents such states efficiently. At the critical point, however, the chain is gapless and is described by a conformal field theory (CFT), and the area law is violated *logarithmically*. For an open chain with a central cut the Calabrese–Cardy result [27] gives

$$S(L)\Big|_{h=h_c} \;=\; \frac{c}{6}\,\ln L \;+\; \text{const}, \tag{3.27}$$

where c is the central charge of the underlying CFT. The "critical entropy" is simply S evaluated at $h = h_c$: unlike its gapped counterpart it grows without bound — logarithmically — with system size, which is the entanglement fingerprint of a one-dimensional critical point.

The DMRG data realise this directly. The measured critical entropies

$$S(h_c) \;=\; 0.42,\; 0.49,\; 0.55,\; 0.61 \qquad \text{for} \qquad L = 16,\, 32,\, 64,\, 128, \tag{3.28}$$

grow logarithmically, and a linear fit of $S(h_c)$ against $\ln L$ (Figure 3.13) has slope $c/6$, yielding

$$c \;\approx\; 0.52, \tag{3.29}$$

in agreement with the exact Ising value $c = \frac{1}{2}$ — the central charge of a single free Majorana fermion. This independently confirms that the transition belongs to the (1+1)D Ising universality class [25], without any reference to the order parameter or the energy.

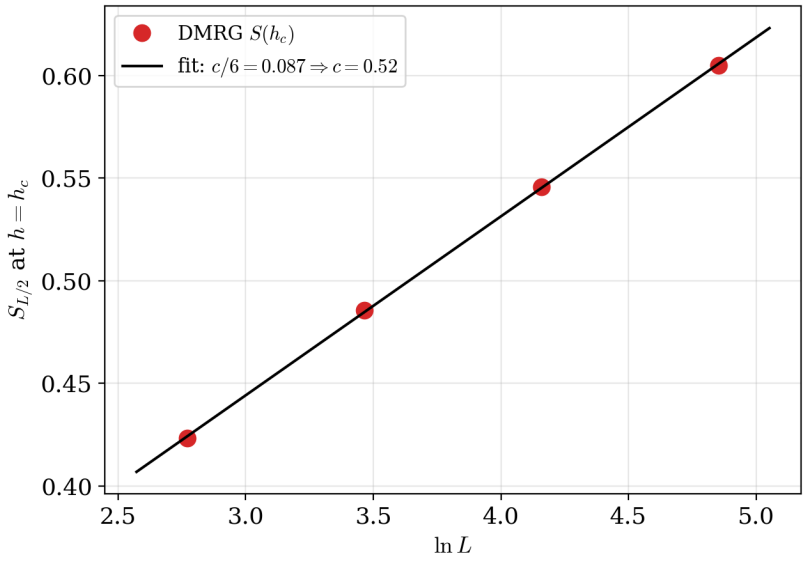


Figure 3.13: Critical entanglement entropy. At $h = h_c$ the half-chain entropy grows as $S = (c/6)\ln L + \text{const}$ (Eq. 3.27); the slope of the DMRG data gives a central charge $c \approx 0.52$, consistent with the exact Ising value $c = \frac{1}{2}$.

3.7.5 Synthesis

Three diagnostics, each extracted from the DMRG ground states and each benchmarked against an exact result, converge on a single conclusion. The order parameter vanishes with exponent $\beta = \frac{1}{8}$; the energy curvature diverges logarithmically ($\alpha = 0$); and the critical entanglement entropy grows with a coefficient fixing the central charge $c = \frac{1}{2}$. Together they identify a continuous quantum phase transition at $h_c/J = 1$ in the (1+1)D Ising universality class. Crucially, every one of these singular features is absent at finite L, where all observables are smooth, and appears only in the thermodynamic limit. The finite-size data therefore make explicit the rigorous distinction — the same one formalised by the free-energy non-analyticity criterion of Section 1.5 — between a smooth crossover and a true phase transition.

Finally, the calculation confirms the practical message of Section 3.6: the bond dimension required for high accuracy stays small (constant away from criticality, growing only logarithmically at the critical point), so the cost of the DMRG is controlled by the entanglement of the target state rather than by the exponentially large dimension of the Hilbert space. The one-dimensional chain is thus both a stringent validation of the method and a clean illustration of why it succeeds.

Chapter 4

Infinite Time-Evolving Block Decimation (iTEBD)

The DMRG of the previous chapter is variational: it searches the manifold of bond-dimension-D matrix product states for the state of lowest energy by repeated local diagonalisation. An entirely different family of algorithms instead obtains the ground state by **imaginary-time evolution**: starting from an arbitrary initial state, one repeatedly applies the operator $e^{-\tau \hat{H}}$ with $\tau > 0$ until all but the ground-state component have decayed. The **Infinite Time-Evolving Block Decimation (iTEBD)** algorithm [12] implements this idea on a translationally invariant infinite chain. It bypasses the sweep structure of DMRG entirely, and its three core ingredients — Trotter decomposition of $e^{-\tau \hat{H}}$ into two-site gates, alternating odd-/even-bond updates, and SVD-based truncation back to bond dimension D — generalise directly to two dimensions as the Simple Update for iPEPS introduced in the next chapter.

4.1 Imaginary-time projection onto the ground state

For any initial state $|\psi\rangle$ with non-zero overlap with the ground state $|g\rangle$, the projection

$$\lim_{\beta \to \infty} e^{-\beta \hat{H}} \, |\psi\rangle \;\propto\; |g\rangle \tag{4.1}$$

holds, since high-energy components are exponentially suppressed. Writing $\hat{H} = \sum_\alpha E_\alpha \, |\alpha\rangle \langle\alpha|$ in the eigenbasis,

$$e^{-\beta \hat{H}} \;=\; \sum_\alpha e^{-\beta E_\alpha} \, |\alpha\rangle \langle\alpha| \;\xrightarrow{\beta \to \infty}\; e^{-\beta E_g} \, |g\rangle \langle g| \,, \tag{4.2}$$

which is, up to a normalisation factor, the projector onto the ground-state subspace. The convergence is exponential in $\beta\Delta$, where Δ is the spectral gap to the first excited state; a finite gap is therefore essential for the method to converge in a tractable number of imaginary-time steps.

4.2 Trotter decomposition

The operator $e^{-\beta\hat{H}}$ cannot be applied as a single MPO without re-introducing the exponential cost that the tensor-network representation is designed to avoid. The standard remedy is to split the Hamiltonian into a sum of nearest-neighbour bond operators,

$$\hat{H} \;=\; \sum_{\ell} \hat{h}_{\ell} \;=\; \hat{H}_o + \hat{H}_e, \tag{4.3}$$

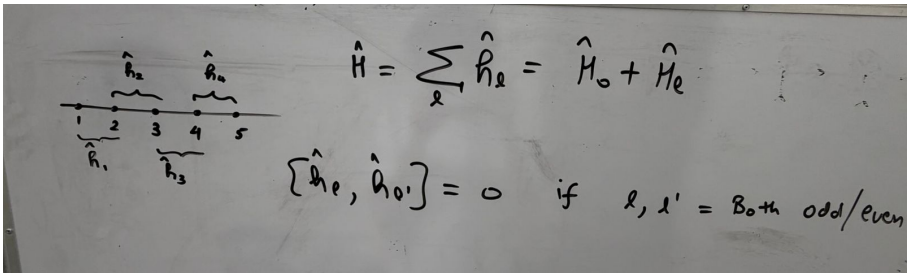


Figure 4.1: Decomposition of the nearest-neighbour Hamiltonian $\hat{H} = \sum_{\ell} \hat{h}_{\ell} = \hat{H}_o + \hat{H}_e$ into odd-bond and even-bond sublattices for iTEBD. Bond operators $\hat{h}_{\ell}$ and $\hat{h}_{\ell'}$ belonging to the same parity act on disjoint site pairs and therefore commute, $[\hat{h}_{\ell}, \hat{h}_{\ell'}] = 0$, allowing their joint exponential to factorise exactly into independent two-site gates.

where $\hat{h}_{\ell}$ acts on the bond connecting sites ℓ and $\ell+1$, $\hat{H}_o = \sum_{\ell \text{ odd}} \hat{h}_{\ell}$, and $\hat{H}_e = \sum_{\ell \text{ even}} \hat{h}_{\ell}$. Different bond operators of the same parity act on disjoint pairs of sites, so they mutually commute:

$$[\hat{h}_{\ell},\, \hat{h}_{\ell'}] \;=\; 0 \quad \text{if } \ell, \ell' \text{ are both odd or both even.} \tag{4.4}$$

The operators $\hat{H}_o$ and $\hat{H}_e$ do *not* commute with each other — they share lattice sites — and this is the source of all Trotter error.

The imaginary-time interval β is divided into N slices of width $\tau = \beta/N$,

$$\beta \;=\; \tau N, \tag{4.5}$$

and the full evolution is written as

$$e^{-\beta\hat{H}} \;=\; \left[e^{-\tau\hat{H}}\right]^N \;=\; \left[e^{-\tau(\hat{H}_o+\hat{H}_e)}\right]^N. \tag{4.6}$$

Expanding each slice using the Baker–Campbell–Hausdorff formula gives the **first-order Trotter approximation**,

$$e^{-\tau(\hat{H}_o+\hat{H}_e)} \;\simeq\; e^{-\tau\hat{H}_o}\, e^{-\tau\hat{H}_e} \;+\; \mathcal{O}(\tau^2), \tag{4.7}$$

or the more accurate **second-order (symmetric) Trotter–Suzuki approximation**,

$$e^{-\tau(\hat{H}_o+\hat{H}_e)} \;\simeq\; e^{-\frac{\tau}{2}\hat{H}_o}\, e^{-\tau\hat{H}_e}\, e^{-\frac{\tau}{2}\hat{H}_o} \;+\; \mathcal{O}(\tau^3), \tag{4.8}$$

in which the error per slice is one power of τ smaller at the cost of a single extra gate factor. The total Trotter error over the full N-slice product is $\mathcal{O}(N\,\tau^p) = \mathcal{O}(\beta\,\tau^{p-1})$ for the pth-order scheme, so reducing τ at fixed β improves accuracy.

Because of the commutativity (4.4), each exponential of $\hat{H}_o$ or $\hat{H}_e$ factorises *exactly* — without further Trotter error — into a product of mutually commuting two-site gates:

$$e^{-\tau\hat{H}_o} \;=\; e^{-\tau\hat{h}_1}\, e^{-\tau\hat{h}_3} \cdots e^{-\tau\hat{h}_{L-1}} \;\equiv\; \hat{U}_1\,\hat{U}_3 \cdots \hat{U}_{L-1}, \tag{4.9}$$

$$e^{-\tau\hat{H}_e} \;=\; e^{-\tau\hat{h}_2}\, e^{-\tau\hat{h}_4} \cdots e^{-\tau\hat{h}_L} \;\equiv\; \hat{U}_2\,\hat{U}_4 \cdots \hat{U}_L. \tag{4.10}$$

Proof that the same-parity exponential is an exact product of two-site gates

The Trotter splitting (4.7)–(4.8) introduces an approximation error because $\hat{H}_o$ and $\hat{H}_e$ do not commute. Once the split is made, however, the factorisation of $e^{-\tau\hat{H}_o}$ and $e^{-\tau\hat{H}_e}$ into products of independent two-site gates is *exact* — no further approximation is involved. We establish this here using the Baker–Campbell–Hausdorff (BCH) formula.

BCH lemma. For any two operators X and Y,

$$e^{X+Y} = e^X\, e^Y\, \exp\!\Big(-\tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) - \cdots\Big). \tag{4.11}$$

Every correction term in (4.11) contains at least one nested commutator of X and Y. Therefore, if $[X,Y]=0$, the entire BCH tail vanishes and

$$[X,Y]=0 \implies e^{X+Y} = e^X\, e^Y \quad \text{(exactly)}. \tag{4.12}$$

Application to $\hat{H}_o$. Write the odd-sublattice Hamiltonian as a finite sum of bond operators,

$$\hat{H}_o = \sum_{\ell\,\text{odd}} \hat{h}_\ell = \hat{h}_1 + \hat{h}_3 + \hat{h}_5 + \cdots \tag{4.13}$$

Each $\hat{h}_\ell$ acts non-trivially only on the pair of sites $(\ell, \ell+1)$. For two distinct odd bonds $\ell \neq \ell'$, the pairs $\{\ell, \ell{+}1\}$ and $\{\ell', \ell'{+}1\}$ are *disjoint*. Because $\hat{h}_\ell$ acts as the identity on every site not in $\{\ell, \ell{+}1\}$, the two operators necessarily commute:

$$[\hat{h}_\ell,\, \hat{h}_{\ell'}] = 0 \qquad (\ell, \ell' \text{ both odd},\ \ell \neq \ell'). \tag{4.14}$$

Applying (4.12) repeatedly — first to the pair $(\hat{h}_1, \hat{h}_3)$, then absorbing $\hat{h}_5$, and so on — gives

$$\begin{aligned} e^{-\tau \hat{H}_o} &= e^{-\tau(\hat{h}_1 + \hat{h}_3 + \hat{h}_5 + \cdots)} \\ &= e^{-\tau \hat{h}_1}\, e^{-\tau \hat{h}_3}\, e^{-\tau \hat{h}_5} \cdots \equiv \hat{U}_1\, \hat{U}_3\, \hat{U}_5 \cdots , \end{aligned} \tag{4.15}$$

where $\hat{U}_\ell := e^{-\tau \hat{h}_\ell}$ is a unitary (imaginary-time: contraction) operator of dimension $d^2 \times d^2$ acting on sites ℓ and $\ell{+}1$ only. Each factor in the product (4.15) commutes with all others, so the ordering is immaterial and the product itself is exact. The identical argument applies to $\hat{H}_e$. The Trotter error therefore resides *entirely* in the splitting $\hat{H} = \hat{H}_o + \hat{H}_e$; once that split is accepted, the two-site gate layers (4.9)–(4.10) follow without any further approximation.

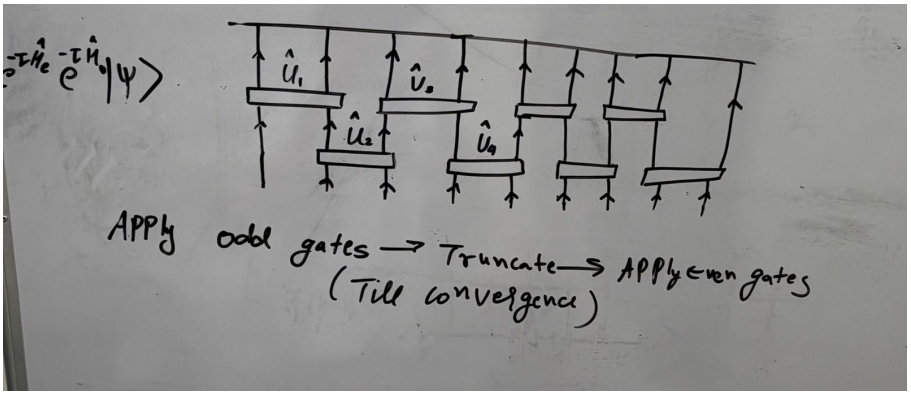


Figure 4.2: iTEBD gate-layer structure for one imaginary-time Trotter step. A layer of odd-bond gates $\hat{U}_1, \hat{U}_3, \ldots$ is applied simultaneously; the resulting bond dimension is truncated back to D via SVD ("Till convergence"). A layer of even-bond gates $\hat{U}_2, \hat{U}_4, \ldots$ follows in the same manner. Together, the two layers implement one first-order Trotter slice $e^{-\tau \hat{H}_e} e^{-\tau \hat{H}_o}$ of the imaginary-time projector.

Each $\hat{U}_\ell$ acts only on the local Hilbert space of two adjacent sites and has dimension $d^2 \times d^2$. Applying $e^{-\tau \hat{H}_o}$ to the chain therefore amounts to applying a *layer* of two-site gates on every odd bond, after which a layer on every even bond applies $e^{-\tau \hat{H}_e}$. Successive layers can be assembled into a matrix product operator whose virtual bonds have dimension $D_{W,e} = 1$ on the even links (no gate spans those) and $D_{W,o} = d^2$ on the odd links (each odd-bond gate occupies one).

4.3 The infinite-chain algorithm

The construction so far works for any chain of length L. A drastic simplification occurs in the thermodynamic limit $L \to \infty$, where translational invariance can be exploited explicitly. With a **two-site unit cell** (M_o, M_e), the entire chain is parameterised by just these two site tensors,

$$\cdots\, M_o\, M_e\, M_o\, M_e\, M_o\, M_e\, \cdots\,,$$

with no left- or right-canonical normalisation imposed on individual tensors. (Vidal's original presentation organises the same data into a Γ–Λ form which makes the canonical form manifest; we work directly with the M tensors throughout this thesis.)

One iTEBD sweep consists of two steps.

Step 1 — odd-bond update. The odd-bond gate $\hat{U}_o = e^{-\tau \hat{h}_o}$ acts on the pair (M_o, M_e). Contracting the gate into the pair gives a single four-index tensor

$$\Theta^{\sigma_o' \sigma_e'}_{\alpha\beta} \;=\; \sum_{\sigma_o, \sigma_e, \gamma} \Big[\hat{U}_o\Big]^{\sigma_o' \sigma_e'}_{\sigma_o \sigma_e} \Big[M_o\Big]^{\sigma_o}_{\alpha\gamma} \Big[M_e\Big]^{\sigma_e}_{\gamma\beta}, \tag{4.16}$$

which is reshaped to an $(Dd) \times (Dd)$ matrix $\Theta_{(\alpha\sigma_o'),\,(\sigma_e'\beta)}$ and singular-value-decomposed,

$$\Theta \;=\; U\, S\, V^{\dagger}. \tag{4.17}$$

Truncating the singular values S to the largest D recovers two updated site tensors $\tilde{M}_o \propto U\sqrt{S}$ and $\tilde{M}_e \propto \sqrt{S}\, V^{\dagger}$ of bond dimension D. This **truncation step** is what keeps the algorithm tractable: without it, the bond dimension would grow by a factor of d with every gate application.

Step 2 — even-bond update. The even-bond gate $\hat{U}_e = e^{-\tau \hat{h}_e}$ is now applied across the *other* bond of the two-site cell — i.e. to the pair $(\tilde{M}_e, \tilde{M}_o)$ formed by an even site and its right-neighbour odd site. The same contract–SVD–truncate procedure produces refreshed tensors $\tilde{\tilde{M}}_e, \tilde{\tilde{M}}_o$.

Together, the two steps implement one (first-order) Trotter step of the imaginary-time evolution on the infinite chain. Iterating Steps 1 and 2 until the energy density per site stops changing within a chosen tolerance yields the converged variational approximation to the ground state. In effect, iTEBD is a **power method** for the operator $e^{-\tau \hat{H}_e}\, e^{-\tau \hat{H}_o}$: every iteration multiplies the state by one more power of the Trotterised projector (4.7), and the ground state is the (unique, for a gapped Hamiltonian) eigenvector of dominant magnitude.

Two distinct sources of error control the final accuracy. The **Trotter error**, $\mathcal{O}(\beta\, \tau^{p-1})$ for

the pth-order splitting, is reduced by shrinking τ. The **truncation error** introduced at each SVD — the sum of discarded squared singular values — is reduced by increasing the bond dimension D. Both must be controlled to obtain accurate ground-state observables, and in practice one converges with respect to each in turn.

Chapter 5

Infinite Projected Entangled-Pair States (iPEPS)

The DMRG of Chapter 3 and the iTEBD of Chapter 4 are both restricted to one-dimensional geometries. In two dimensions, neither is satisfactory: DMRG on a "snake-MPS" of a 2D lattice can certainly be done, but the bond dimension required to capture the entanglement of even a gapped 2D ground state grows exponentially with the linear size of the system. This is the fundamental limitation that motivates the present chapter. The **projected entangled-pair state (PEPS)** is the natural two-dimensional generalisation of an MPS, introduced by Verstraete and Cirac in 2004 [13], and its translationally invariant version, **iPEPS**, is the variational ansatz used for all the numerical results in this thesis.

The chapter is in three parts. We first explain why a 2D area law forces the introduction of PEPS, then define the ansatz, and finally describe the two algorithms — **Simple Update (SU)** and **Full Update (FU)** — by which the ground state is obtained via imaginary-time evolution.

5.1 Why DMRG fails in two dimensions: the area-law argument

Two-dimensional DMRG (sometimes called "snake-MPS DMRG") works by mapping the 2D lattice onto a 1D chain along a snake-like path and applying the standard MPS sweep algorithm. The strategy is one of the most accurate methods available for 2D quantum lattice models [15]; it has been used for the 2D Heisenberg model [14], the Kagome lattice [16], and recently for striped order in the 2D Hubbard model [17]. However, the snake construction has a built-in scaling problem that we now make explicit.

An MPS of bond dimension D can encode at most

$$S_{AB}^{\mathrm{MPS}} \;=\; \mathcal{O}(\ln D)$$

units of entanglement entropy across *any single* virtual bond. By contrast, the **two-dimensional area law** for the ground state of a gapped local Hamiltonian on a 2D lattice predicts that the entanglement entropy between a region A and its complement scales with the *length* of the cut,

$$S_{AB}^{\text{2D area law}} \;\sim\; L,$$

where L is the linear extent of the boundary between A and $\bar{A}$. Demanding that a snake-MPS reproduce this scaling forces

$$\ln D \;\overset{!}{\sim}\; L \quad\Longrightarrow\quad D \;\sim\; e^{L}.$$

The bond dimension required for a snake-MPS to faithfully represent a 2D ground state therefore grows **exponentially** with the linear size of the system. This is not a peculiarity of the snake mapping — it is the fundamental geometric mismatch between a one-dimensional ansatz and a two-dimensional area law. All of the entanglement between two halves of the 2D system has to flow through whichever single virtual MPS bond the snake places at the cut, and that one bond is forced to carry an amount of entanglement that scales as the perimeter.

The natural remedy, formalised by Verstraete and Cirac, is to **add more virtual bonds** [13]: associate one virtual leg with every direction in which two nearest-neighbour sites of the 2D lattice are connected, so that the entanglement across an area-L cut is shared among $\mathcal{O}(L)$ bonds rather than being squeezed through a single one. The resulting ansatz is the PEPS.

5.2 Definition of the PEPS ansatz

On a square lattice of size $N_x \times N_y$, the PEPS associates with each site $\ell = (\ell_x, \ell_y)$ a single five-index tensor

$$A^{\sigma_\ell}_{[\ell]\,\alpha\beta\gamma\delta},$$

where σ_ℓ is the physical index of local dimension d and $\alpha, \beta, \gamma, \delta$ are four virtual indices (left, up, down, right) of bond dimension D that are contracted with the matching virtual indices of the four nearest-neighbour tensors. The wavefunction is obtained by summing

over all virtual bonds,

$$|\psi\rangle \;=\; \sum_{\vec{\sigma}} |\vec{\sigma}\rangle \prod_{\ell} A^{\sigma_\ell}_{[\ell]}, \tag{5.1}$$

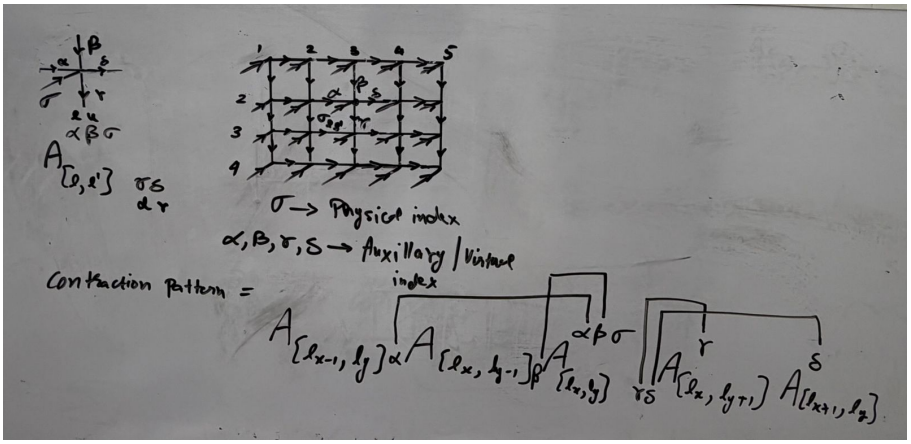


Figure 5.1: iPEPS wavefunction on the infinite square lattice. Each site $\ell = (\ell_x, \ell_y)$ hosts a rank-5 tensor $A^{\sigma_\ell}_{[\ell]\,\alpha\beta\gamma\delta}$ with one physical index σ (local Hilbert space dimension d) and four virtual indices $\alpha, \beta, \gamma, \delta$ of bond dimension D, each contracted with the corresponding virtual index of the nearest-neighbour tensor. The wavefunction is the tensor-network contraction over all virtual bonds on the infinite lattice.

where the product implicitly contracts each virtual index at (ℓ_x, ℓ_y) with the matching one at $(\ell_x \pm 1, \ell_y)$ or $(\ell_x, \ell_y \pm 1)$. The number of variational parameters is

$$\#\text{params} \;=\; \mathcal{O}(d\, D^4\, N_x\, N_y),$$

in contrast to the $\mathcal{O}(d\, D^2\, N_x\, N_y)$ of a snake-MPS — the extra factor of D^2 reflects the two additional virtual legs at every site.

In the **infinite** version of PEPS (iPEPS), the lattice is taken to be infinite, $N_x, N_y \to \infty$, and translational invariance is exploited explicitly by requiring the site tensors to repeat with a small unit cell — typically 1×1 or 2×2. This is the direct two-dimensional generalisation of the two-site unit cell used in iTEBD, and is the variant used in this thesis.

The name "projected entangled-pair state" originates in a constructive interpretation that generalises the AKLT chain. With each site ℓ associate four auxiliary virtual particles labelled by indices in the four lattice directions,

$$|\alpha\beta\gamma\delta\rangle_\ell \;=\; |\alpha\rangle^{l}_{\ell}\; |\beta\rangle^{u}_{\ell}\; |\gamma\rangle^{d}_{\ell}\; |\delta\rangle^{r}_{\ell}\,.$$

For every nearest-neighbour bond $\langle \ell, \ell' \rangle$, define a maximally entangled pair of the two

auxiliary particles facing across the bond,

$$|\mathrm{EP}\rangle_{\ell,\ell'} \;=\; \sum_{\gamma=1}^{D} |\gamma\rangle_{\ell}\, |\gamma\rangle_{\ell'}\,, \tag{5.2}$$

and on each site a *projector* from the four auxiliary particles onto the physical state,

$$\hat{P}_{\ell} \;=\; \sum_{\sigma,\alpha\beta\gamma\delta} |\sigma\rangle_{\ell}\; A^{\sigma}_{[\ell]\,\alpha\beta\gamma\delta}\; \langle\alpha\beta\gamma\delta|\,. \tag{5.3}$$

The PEPS is then

$$|\psi\rangle \;=\; \Big(\prod_{\ell} \hat{P}_{\ell}\Big)\Big(\prod_{\langle\ell,\ell'\rangle} |\mathrm{EP}\rangle_{\ell,\ell'}\Big), \tag{5.4}$$

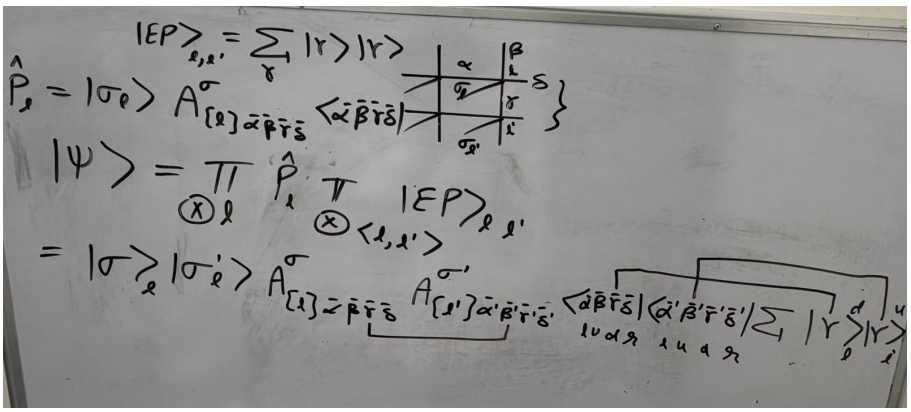


Figure 5.2: Projected entangled-pair construction of the PEPS. For each nearest-neighbour bond $\langle\ell,\ell'\rangle$ a maximally entangled pair $|\mathrm{EP}\rangle_{\ell,\ell'} = \sum_{\gamma=1}^{D} |\gamma\rangle_{\ell}|\gamma\rangle_{\ell'}$ is placed on the virtual particles facing across that bond. A local projector $\hat{P}_{\ell} = \sum_{\sigma,\alpha\beta\gamma\delta} |\sigma\rangle_{\ell} A^{\sigma}_{[\ell]\alpha\beta\gamma\delta} \langle\alpha\beta\gamma\delta|$ at each site maps the four virtual particles onto the physical Hilbert space, yielding $|\psi\rangle = \prod_{\ell} \hat{P}_{\ell} \prod_{\langle\ell,\ell'\rangle} |\mathrm{EP}\rangle_{\ell,\ell'}$.

i.e. a state of entangled auxiliary pairs *projected* onto the physical Hilbert space at every site — hence the name.

5.3 Properties of PEPS

Three structural properties of the ansatz are central to its role in this thesis [18].

(i) PEPS satisfy the 2D area law. Cutting a region A of perimeter L_A from a square-lattice PEPS severs $\mathcal{O}(L_A)$ virtual bonds rather than a single one. Each bond carries at most $\ln D$ units of entanglement, so

$$S_{AB}^{\mathrm{PEPS}} \;\sim\; \mathcal{O}(L_A \ln D). \tag{5.5}$$

A PEPS of *constant* bond dimension D already saturates the 2D area law, in stark contrast to the $D \sim e^L$ required by a snake-MPS. This is the single most important reason for using PEPS in two dimensions.

(ii) PEPS can support polynomially-decaying correlations. Unlike one-dimensional MPS, whose two-point correlations decay exponentially with distance for any finite D, two-dimensional PEPS can encode algebraic correlations — making them appropriate for critical points such as the 2D TFIM transition.

(iii) Exact contraction is computationally hard. The price paid for the richer structure is that contracting a PEPS to evaluate an expectation value $\langle\psi|\,\hat{O}\,|\psi\rangle$ is #P-hard in general. Every practical PEPS algorithm therefore relies on an **approximate** contraction scheme; in this thesis we use the Corner Transfer Matrix Renormalisation Group (CTMRG), characterised by a separate environment bond dimension χ and elaborated in the next chapter.

5.4 Ground-state search by imaginary-time evolution

Given the PEPS ansatz, how does one optimise its tensors to approximate the ground state of a target Hamiltonian $\hat{H}$? The strategy we use — and the one supported by our GPU implementation — is **imaginary-time evolution**, in direct analogy with iTEBD: a Trotter decomposition of $e^{-\tau\hat{H}}$ into local two-site gates, followed at every step by a truncation back to bond dimension D.

On a square lattice the nearest-neighbour Hamiltonian splits naturally into four mutually commuting subsets, distinguished by parity and orientation [19]:

$$\hat{H} \;=\; \hat{H}_{eh} + \hat{H}_{oh} + \hat{H}_{ev} + \hat{H}_{ov}, \tag{5.6}$$

where e/o stands for even/odd and h/v for horizontal/vertical. Each of the four subsets is itself a sum of bond operators acting on disjoint pairs of sites and therefore consists of mutually commuting terms. A first-order Trotter decomposition of one imaginary-time slice τ reads

$$e^{-\tau\hat{H}} \;\simeq\; e^{-\tau\hat{H}_{eh}}\, e^{-\tau\hat{H}_{oh}}\, e^{-\tau\hat{H}_{ev}}\, e^{-\tau\hat{H}_{ov}} \;+\; \mathcal{O}(\tau^2). \tag{5.7}$$

Each factor is, by the same factorisation argument as in iTEBD, an exact product of mutually commuting two-site gates on the corresponding bonds. Applying one factor — say $e^{-\tau\hat{H}_{eh}}$ — to all even-horizontal bonds of the PEPS produces an updated state which

we denote schematically as

$$|\psi_B\rangle \;=\; \prod_{\langle \ell,\ell' \rangle \,\in\, eh} e^{-\tau \hat{h}_{\ell\ell'}}\; |\psi_A\rangle\,. \tag{5.8}$$

Each two-site gate, contracted into the pair of PEPS tensors it spans, *inflates* the bond between them: a previously D-dimensional virtual bond becomes a bond of dimension Dd^2. Without truncation, every iteration of the full slice (5.7) would multiply each bond dimension by d^8, and the algorithm would be unusable after a handful of slices.

5.5 Truncation: Simple Update and Full Update

Two distinct schemes are used in this thesis to truncate the bond dimension back to D after each gate application; they are the two methods responsible for the two families of datasets in our results chapter.

Simple Update (SU). The simplest scheme works one bond at a time. The pair of PEPS tensors sharing the active bond is contracted together with the gate; the resulting object is reshaped into a matrix and singular-value-decomposed,

$$\Theta \;=\; U\,S\,V^{\dagger},$$

and only the largest D singular values are retained, yielding updated tensors $\tilde{A}_\ell, \tilde{A}_{\ell'}$. The procedure is the direct two-dimensional analogue of Step 1 of the iTEBD algorithm of Chapter 4. SU is fast — the dominant cost per bond is a single SVD — but the truncation is purely local: the directions discarded are those of smallest singular value *with respect to the bond itself*, regardless of how the rest of the PEPS depends on the truncated indices. In effect, SU treats the environment of every bond as if it were a product state, which it is not.

Full Update (FU). A more accurate but considerably more expensive scheme treats the truncation as a global variational problem. After the bond-inflating gate has been applied, one seeks a PEPS $|\psi'\rangle$ of the target bond dimension D that is as close as possible to the time-evolved state $\left|\tilde{\psi}\right\rangle$ of bond dimension Dd^2, by minimising the cost function

$$K \;=\; \left\| \,|\psi'\rangle - \left|\tilde{\psi}\right\rangle \right\|^2 \;=\; \langle\psi'|\psi'\rangle + \left\langle\tilde{\psi}\middle|\tilde{\psi}\right\rangle - 2\,\mathrm{Re}\,\left\langle\psi'\middle|\tilde{\psi}\right\rangle \tag{5.9}$$

one tensor at a time. Setting the derivative with respect to a single conjugate site tensor $A'^{\dagger}_{[\ell]}$ to zero,

$$0 \;=\; \frac{\partial K}{\partial A'^{\dagger}_{[\ell]}} \quad\Longrightarrow\quad N_\ell\, A'_\ell \;=\; W_\ell, \tag{5.10}$$

yields a linear system whose solution updates the tensor at site ℓ. Here N_ℓ is the **norm tensor**, obtained by contracting the double-layer network $\langle\psi'|\psi'\rangle$ with the active tensor and its conjugate removed, and W_ℓ is the corresponding contraction of $\left\langle\psi'\middle|\tilde{\psi}\right\rangle$ with only $A'^{\dagger}_{[\ell]}$ removed. Both N_ℓ and W_ℓ are obtained from the double-layer PEPS network — the two-layer geometry produced by stacking $|\psi'\rangle$ above its conjugate $\langle\psi'|$ — by approximate contraction with environment bond dimension χ (the CTMRG of the next chapter, in this thesis). By incorporating the full PEPS environment into the truncation, FU produces a substantially more accurate ground-state energy at fixed bond dimension D, at the cost of the repeated approximate contraction of the double-layer network — which dominates the runtime of any practical Full Update calculation and is the principal reason that the GPU acceleration described in a later chapter is essential.

A practical workflow combines the two: a fast SU sweep first brings the PEPS close to the ground state, and a Full Update polish at the desired final accuracy produces production-quality results. Both protocols are implemented in our code and benchmarked against each other in the results chapter.

Chapter 6

Corner Transfer Matrix Renormalisation Group

The Full Update truncation introduced in Chapter 5 requires, at every gate application, an approximate evaluation of contractions of the type $\langle\psi'|\psi'\rangle$ and $\left\langle\psi'\middle|\tilde{\psi}\right\rangle$ on the infinite square lattice. Since exact contraction of an iPEPS network is #P-hard, this evaluation can only ever be performed approximately, and the quality of the truncation — and ultimately of the ground-state energy — is set by the quality of this approximation. The standard scheme used in this thesis is the **Corner Transfer Matrix Renormalisation Group (CTMRG)**, originally proposed by Nishino and Okunishi [22] as a tensor-network adaptation of Baxter's classical-statistical-mechanics corner transfer matrices [20, 21], and adapted to the iPEPS setting by Orús and Vidal [23]. We follow the conventions of the pedagogical review by Bruognolo, Li and von Delft [24], whose notation we adopt below.

6.1 Philosophy of coarse-graining

The fundamental obstacle is that the network we want to contract — the double-layer iPEPS that represents $\langle\psi|\,\hat{O}\,|\psi\rangle$ — is infinite in both lattice directions. Any practical evaluation must therefore replace the infinite environment by a *finite* representation, and this is exactly what every renormalisation-group method does: integrate out a portion of the degrees of freedom, retain only those effective parameters that govern the action of the integrated-out part on what remains, and iterate.

In CTMRG the "kept" piece is a small **window** — the unit cell of the iPEPS, here taken as either 1×1 or 2×2. Everything outside the window — the **environment** — is replaced by a small set of tensors that summarise its action on the window's boundary. The environment tensors are not obtained in one shot; they are built up self-consistently,

by repeatedly enlarging the active region by one extra row or column of unit cells and then *compressing* the boundary back to its original size. After sufficiently many such iterations, the environment converges to a fixed point that encodes, to controlled accuracy, the contribution of the entire infinite half-plane lying outside the window.

The structural parallel with Wilson's numerical renormalisation group is direct: Wilson keeps the lowest-energy states from each shell of an integrated-out lattice; CTMRG keeps the dominant **environment indices** of dimension χ — the **environment bond dimension** — which encode the entanglement structure of the half-plane to the left, right, above or below the window. The parameter χ controls the accuracy of the environmental approximation in exactly the same way that D controls the accuracy of the iPEPS ansatz itself, and both must be converged before any iPEPS result can be trusted.

6.2 The CTM ansatz

For a 2×2 unit cell of the iPEPS embedded in the infinite lattice, the double-layer network around the active window is replaced by an effective representation built from twelve environment tensors and four bulk tensors [24]:

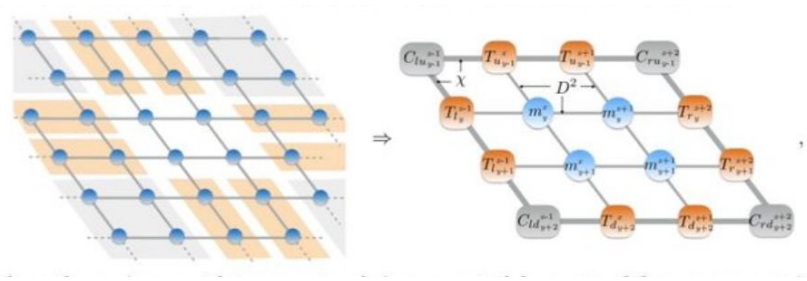


Figure 6.1: CTM ansatz for a 2×2 iPEPS unit cell embedded in the infinite lattice. The infinite environment surrounding the active window (left, shaded orange) is replaced by a compact set of 16 tensors (right): four $\chi \times \chi$ corner matrices $C_{lu}, C_{ld}, C_{ru}, C_{rd}$; eight rank-3 transfer tensors T_l, T_r, T_u, T_d (two of each type, one per row/column of the unit cell) with two χ-legs and one D^2-leg; and four double-layer bulk tensors $M_{[\ell]}$ of leg dimension D^2 obtained by tracing out the physical index of each PEPS tensor against its conjugate(Image taken from [24]).

- **Four corner matrices** $C_{lu}, C_{ld}, C_{ru}, C_{rd}$, where the subscripts denote spatial location (left/right, up/down). Each is a rank-two tensor of dimension $\chi \times \chi$.
- **Eight transfer tensors** T_l, T_r, T_u, T_d — two of each type, one for each row or column of the unit cell. Each is a rank-three tensor with two environment legs of dimension χ and one bulk leg of dimension D^2 pointing into the active window.
- **Four bulk tensors** M (one for each site of the 2×2 unit cell), each carrying dimension D^2 on all four legs. The bulk tensor at site ℓ is the **double-layer tensor**

obtained by contracting the physical index between a PEPS tensor and its conjugate,

$$M_{[\ell]} \;=\; \sum_{\sigma} A^{\sigma}_{[\ell]} \otimes \left(A^{\sigma}_{[\ell]}\right)^{\dagger}, \tag{6.1}$$

so that each of its four virtual legs carries dimension D^2.

The required bond-dimension hierarchy is

$$\chi \;\geq\; D^2. \tag{6.2}$$

A choice of $\chi < D^2$ would discard information at every renormalisation step and is generically uncontrolled; values somewhat above D^2 are required to achieve convergence near criticality.

This ansatz makes the central CTMRG approximation explicit: an infinite tensor network has been replaced by a finite, bounded representation involving exactly $4 + 8 + 4 = 16$ tensors, of which only the $4 + 8 = 12$ environment tensors carry information about the infinite half-planes outside the active window. The price for this drastic compression is that the environment tensors must themselves be obtained by a self-consistent iterative procedure — the directional coarse-graining moves of CTMRG.

6.3 Directional coarse-graining moves

A single **directional move** updates the environment in one of the four lattice directions. Consider, for definiteness, a move to the left. The move consists of three steps [24]:

(i) Insertion. An extra column of unit-cell tensors — one column of M's plus the corresponding boundary T's on the top and bottom and corner C's at the ends — is inserted between the active window and the existing left environment. The active region has now grown by one column.

(ii) Absorption. The newly inserted column is contracted with the left environment tensors C_{lu}, T_l, C_{ld}. This produces an enlarged set of "left-environment" tensors whose environment-leg dimension has grown from χ to χD^2, since each transfer tensor has a D^2 leg that has been absorbed.

(iii) Renormalisation. The enlarged tensors of environment-leg dimension χD^2 are projected back down to dimension χ using an isometric projector built from the dominant singular vectors of an appropriate boundary contraction. The result is an updated set of left-environment tensors C'_{lu}, T'_l, C'_{ld}, again of dimension χ.

Steps (ii)–(iii) are repeated for each row or column of the inserted unit cell — typically twice for a 2×2 unit cell — until the inserted column has been fully digested into the left environment. The whole procedure is then repeated for the right, up, and down directions. One **CTMRG sweep** consists of one such directional move in each of the four lattice directions.

The choice of projector in step (iii) is the most delicate part of the algorithm. Because an iPEPS does not admit an exact canonical form — in contrast to an MPS, which can always be truncated optimally with a single SVD — there is no unique "best" projector. Several schemes are in use [24]; the most widely used builds a directional projector from the SVD of one corner of the enlarged environment. The schemes differ in how much of the environment is fed into the construction of the projector, and consequently in their accuracy for inhomogeneous or critical systems.

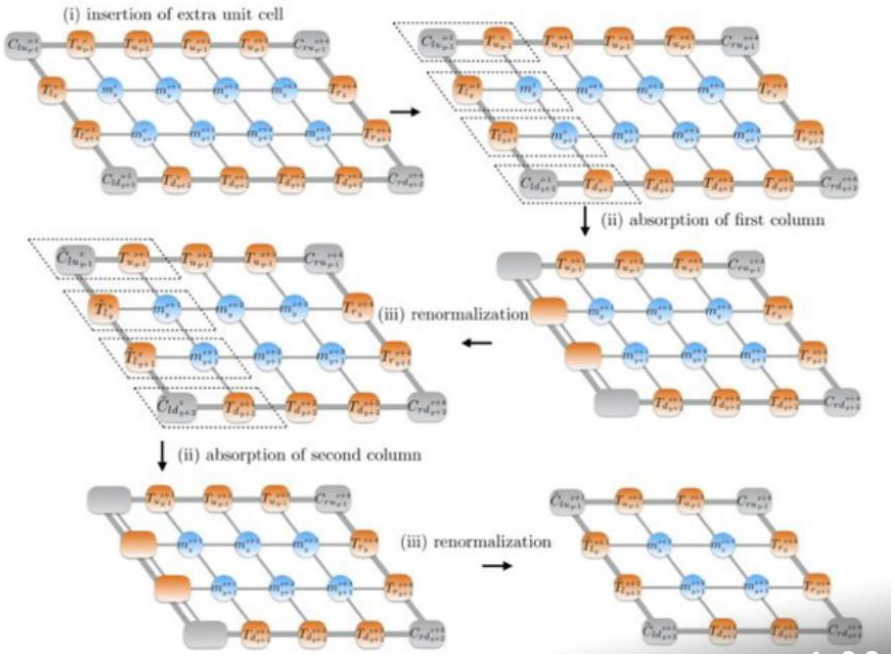


Figure 6.2: CTMRG directional coarse-graining move (left direction, 2×2 unit cell, first column). *Top row:* an extra column of unit-cell tensors is inserted between the active window and the existing left environment, then absorbed into the corner and transfer matrices C_{lu}, T_l, C_{ld}, enlarging their environment legs from χ to χD^2. *Bottom row:* the enlarged tensors are projected back to dimension χ by an isometric projector built from the dominant singular vectors of the boundary contraction, yielding updated tensors C'_{lu}, T'_l, C'_{ld}.(Image Taken from [24])

6.4 Convergence

CTMRG is run as a fixed-point iteration: sweeps are repeated until the environment tensors stop changing within a chosen tolerance, typically measured on the singular-value

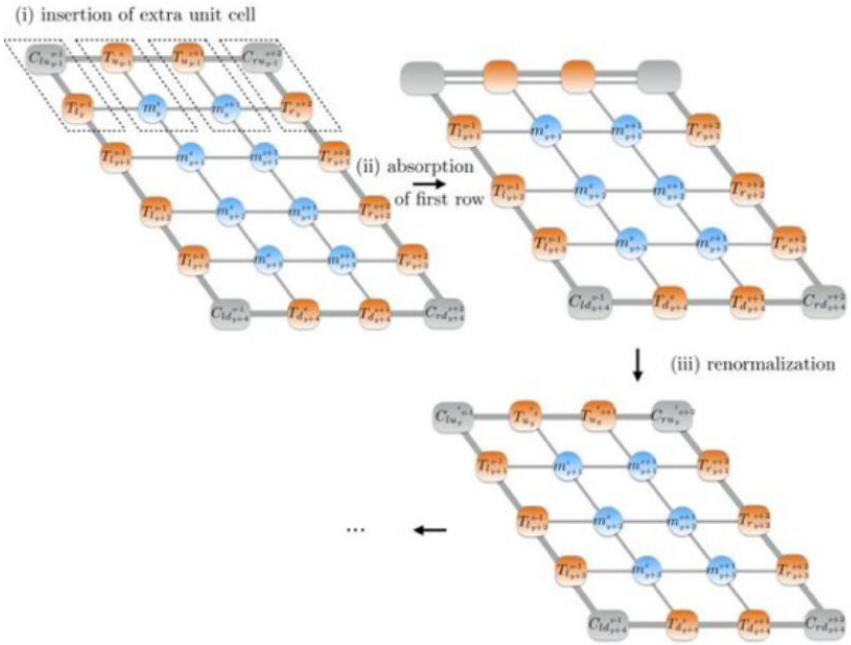


Figure 6.3: CTMRG directional coarse-graining move (left direction, second column). After absorbing the first column of the 2×2 unit cell (previous panel), the procedure repeats: a second absorption step enlarges the environment-leg dimension from χ to χD^2 once more, and a second renormalisation step compresses it back to χ using a freshly constructed isometric projector. The full left-direction move (two absorptions plus two renormalisations) is then repeated for the right, up, and down directions to complete one full CTMRG sweep.

spectra of the corner matrices C. The number of sweeps required depends sharply on the underlying physics:

- For a **gapped** system away from any quantum critical point, the correlation length is finite, the environment depends only on a finite neighbourhood of the active window, and a handful of sweeps — typically of order ten — suffices.
- For a **critical** system, the correlation length diverges, the environment retains memory of arbitrarily distant sites, and the number of sweeps required can grow well beyond a hundred. For fixed χ, CTMRG imposes an effective finite correlation length even at criticality, and the resulting biases must be controlled by finite-χ scaling.

The TFIM at the critical field $(h/J)_c \approx 3.044$ studied in this thesis sits precisely in the second regime, and our calculations require both large χ and a large number of CTMRG sweeps.

6.5 Computational cost and the need for GPU acceleration

The CTMRG step, embedded inside the Full Update inner loop, is the computational bottleneck of any iPEPS calculation. Two operations dominate.

(a) Tensor contractions. Each absorption step (ii) contracts environment tensors of leg dimension χ with a column of bulk and transfer tensors carrying D^2 legs. Depending on the order in which the indices are contracted, the most expensive single contraction in a CTMRG sweep scales as

$$T_{\text{absorb}} \;=\; \mathcal{O}\big(\chi^3 D^4\big) \;\text{ or }\; \mathcal{O}\big(\chi^2 D^6\big). \tag{6.3}$$

With $\chi \sim D^2$ this becomes $\mathcal{O}(D^8)$ — a steep scaling that becomes punishing for the moderately large bond dimensions (D up to 12 in this work) required to converge ground-state observables.

(b) Singular value decompositions. The projector construction in step (iii) requires the singular value decomposition of a $\chi D^2 \times \chi D^2$ matrix, costing

$$T_{\text{SVD}} \;=\; \mathcal{O}\big(\chi^3 D^6\big) \tag{6.4}$$

floating-point operations per projector. Several such SVDs are performed per directional move, four directions are needed per sweep, and tens to hundreds of sweeps are needed

per imaginary-time Trotter step. The cost compounds aggressively.

These costs combine in the Full Update inner loop. Every gate application in the Trotter sequence (5.7) calls CTMRG once for the environment and again for the linear-system solve in equation (5.10); a production run involves 10^3–10^4 Trotter steps, each containing four gate factors, each calling a CTMRG inner loop that itself comprises tens of sweeps. The total operation count is dominated by the dense contractions and SVDs above. On a single CPU core, such a calculation runs for days or even weeks; on a multi-core CPU, the parallel speed-up is bounded by the limited parallelism available within a single dense tensor contraction.

The picture changes qualitatively on a GPU. The dense tensor contractions of step (ii) are essentially large matrix multiplications, which are precisely the workload that GPUs are designed for: modern devices sustain tens of teraflops on problems of the relevant size, and the arithmetic intensity of these contractions is high enough to keep the cores fed. The dense SVDs of step (iii), once the matrix dimension χD^2 is sufficiently large (in practice above a few hundred), are likewise well served by GPU linear-algebra backends. Empirically, our implementation of CTMRG on a single consumer GPU yields speed-ups of one to two orders of magnitude over equivalent single-core CPU code for the bond dimensions used in this thesis, transforming production runs that would otherwise take weeks into runs that complete in hours. This is the central practical reason for the GPU acceleration developed for this thesis, without which the bond dimensions and number of Trotter steps that we actually use would not be tractable. The implementation details of the GPU code — choice of library, data layout, kernel fusion, memory management — are the subject of the next chapter.

Chapter 7

Implementation and Numerical Results

The literature chapters described what iPEPS *is* and how the Simple Update and Full Update algorithms *should* work. This chapter describes what was actually built, what was actually run, and what came out. We start with the computational architecture, walk through the lattice and model implementation, list the convergence parameters used in production, present the four production datasets in order of increasing (D, χ) for the SU and FU runs separately, and conclude with discussion — including counterintuitive features and the limitations that bound the work. The complete source code is available on GitHub.[1]

7.1 Computational architecture

The implementation is two Julia files sitting on the CUDA.jl / cuTENSOR / TensorOperations.jl stack: `ipeps_tfim_gpu.jl` (1301 lines), which implements the iPEPS tensors, Simple Update, CTMRG and observables, and `ipeps_fu.jl` (621 lines), which `includes` the first file and adds the Full Update layer on top. Each production sweep is driven by a small script — `benchmarks.jl` for SU, `fu_sweep_chained.jl` for FU — that calls the library and writes one CSV plus one serialised cache per run. Postprocessing and plotting are done in Python via `plot_latest.py`. All four production sweeps were submitted as SLURM jobs to a campus cluster: the SU runs on an NVIDIA H100 (80 GB, partition `gpu_h100_4`), the FU runs on a V100 (32 GB, partition `gpu_v100_2`) since FU at $D \leq 6$ needs far less memory than SU at $D = 12$.

[1] `https://github.com/<your-username>/ipeps-tfim` — replace with the actual repository URL once the code is pushed.

The memory budget is the single most important architectural fact. The four double-layer tensors $M_{[\ell]}$ at $D = 12$ each occupy $144^4 \times 16$ B ≈ 6.9 GB, so the four-tensor resident set is already 27.5 GB before any environment is built. The $\chi D^2 \times \chi D^2 = 5760 \times 5760$ matrix entering each CTMRG projector SVD is another 530 MB, with a comparable workspace from the Jacobi SVD routine `cuSOLVER.gesvdj!`. The peak transient — an operator-inserted double-layer tensor $\sum_{\sigma\sigma'} A^{\sigma} \sigma^z_{\sigma\sigma'} (A^{\sigma'})^\dagger$ inside `expect_2site` — is another 13.7 GB on top of the resident set. The first version of the code OOMed on the 80 GB H100; the production version aggressively frees CUDA buffers as soon as they are dead (`CUDA.unsafe_free!` in a try/catch double-free guard), capping peak resident usage at $\sim$ 45–55 GB. The FU memory budget is much smaller (the four bulk tensors at $D = 6$ are 104 MB total) which is why FU runs comfortably on a V100; at $D = 12$, however, the FU intermediate of shape $[D^2, D^2, rD, D^2, \chi, \chi]$ would be ~ 1 TB, which is what motivates the hard restriction $D_{\mathrm{FU}} \leq 8$, $\chi_{\mathrm{FU}} \leq 60$ enforced as an assertion in `ipeps_fu.jl`.

7.2 Lattice geometry, model, and tensors

The lattice is the infinite square lattice with a 2×2 unit cell — the smallest cell that distinguishes even and odd parities of bonds in both directions. The four sites of the cell are labelled A, B, C, D in the arrangement

$$\begin{array}{cc} A & B \\ C & D \end{array}$$

and the four bonds inside the cell are the two horizontal bonds AB (top), CD (bottom), and the two vertical bonds AC (left), BD (right). Each site carries a rank-five PEPS tensor $A^{\sigma}_{[\ell]\,\alpha\beta\gamma\delta}$ in the conventions of Chapter 5, with physical dimension $d = 2$ and bond dimension D on each of the four virtual legs. In the Julia code the legs are ordered (σ, U, R, D_n, L) — physical, up, right, down, left.

The Hamiltonian is the transverse-field Ising model on this lattice,

$$\hat{H} \;=\; -J \sum_{\langle \ell, \ell' \rangle} \hat{\sigma}^z_\ell \, \hat{\sigma}^z_{\ell'} \;-\; h \sum_{\ell} \hat{\sigma}^x_\ell . \tag{7.1}$$

All calculations use $J = 1$, which sets the energy scale; h is the only free parameter that varies across a sweep. The critical point at which the Z_2 symmetry breaks is at $(h/J)_c = 3.04438(2)$ from QMC [6], against which we benchmark.

There is one subtlety in the implementation worth highlighting. In a pure imaginary-time evolution of a Z_2-symmetric Hamiltonian, the ground state in the ordered phase has spontaneously broken symmetry and the variational manifold has two degenerate solutions

$\pm m^z$. Without explicit symmetry breaking, each h-point converges to a random branch, producing a noisy $m^z(h)$ plot. The Trotter gate is therefore generated from a slightly modified bond Hamiltonian,

$$\hat{h}_{\text{bond}} \;=\; -J\,\sigma^z \otimes \sigma^z \;-\; \tfrac{h}{4}\Big(\sigma^x \otimes I + I \otimes \sigma^x\Big) \;-\; \tfrac{\varepsilon}{4}\Big(\sigma^z \otimes I + I \otimes \sigma^z\Big), \tag{7.2}$$

with $\varepsilon = 10^{-3}$. This explicit field pins every h-point to the $+m^z$ branch without measurably shifting the transition. The reported $\langle\sigma^z\rangle$ values therefore have a consistent sign — except in a few isolated h-points where the residual noise still flips the sign; in those points we report $|\langle\sigma^z\rangle|$.

7.3 Algorithmic pipeline

Each h-point in a Simple-Update sweep follows the four-stage pipeline of Chapters 4, 5, and 6.

Stage 1 — Simple Update. The PEPS tensors and weights are initialised (random Gaussian, normalised), and a five-step $\delta\tau$-schedule

$$(\delta\tau, n_{\text{sweeps}}) \;\in\; \Big\{(10^{-1}, 50),\, (5\times10^{-2}, 100),\, (10^{-2}, 300),\, (5\times10^{-3}, 500),\, (10^{-3}, 500)\Big\}$$

is applied in succession. Each $\delta\tau$ slice generates a two-site gate $\hat{U} = e^{-\delta\tau\,\hat{h}_{\text{bond}}}$ that is applied bond-by-bond in the order $AB \to CD \to AC \to BD$ (one pass per inner iteration), each bond followed by an SVD truncation back to dimension D. This is the four-term Trotter decomposition (5.7) realised as four bond updates per pass. The schedule's monotonic decrease in $\delta\tau$ produces an effective annealing: the early large-$\delta\tau$ passes drive the state into the correct broad region of the variational manifold; the late small-$\delta\tau$ passes converge it to the local minimum.

Stages 2–3 — Double layer + CTMRG. The four PEPS tensors are contracted with their conjugates (eq. (6.1)) to form the four bulk tensors M_A, M_B, M_C, M_D; the CTM environment of corner matrices and transfer tensors is initialised at random (Gaussian, hermitised for the corners); then directional sweeps LEFT → UP → RIGHT → DOWN are iterated. Each direction calls a single function (`_absorb_left_col` and its three siblings) that performs the three-step insert–absorb–renormalise move of Chapter 6: forming the enlarged corner nC_{lu}, building the $\chi D^2 \times \chi D^2$ matrix Ψ, taking its singular value decomposition via the GPU Jacobi routine, and slicing the top χ singular vectors to build the projector. Each absorption normalises its output tensors immediately to avoid magnitude blow-up across many sweeps. The CTMRG loop terminates when the

ℓ^2-distance between successive normalised singular-value spectra of C_{lu} drops below 10^{-4}, capped at 60 sweeps.

Stage 4 — Observables. With the converged environment in hand, the energy is computed as the sum of four bond expectation values plus a transverse field contribution. Each two-site bond expectation value $\langle \sigma_A^z \sigma_B^z \rangle$ is evaluated by SVD-decomposing the bond operator into rank-one components and contracting four enlarged corners of the 2×2 network for each component, which keeps the peak intermediate at 530 MB rather than the $\sim$ 76 GB that the naive row-by-row contraction would produce. The transverse magnetisation $\langle \sigma^x \rangle$ is the average of four one-site insertions; the longitudinal $\langle \sigma^z \rangle$ is the average of the same four sites. The boundary entropy — the von Neumann entanglement entropy of the boundary cut, computed from the squared, normalised singular spectrum of the boundary transfer tensor T_{u1} — is the proper area-law diagnostic.[2] Output is serialised to disk as a `.jls` cache plus a CSV row.

Full Update. For an FU run the pipeline is identical through Stage 3 — there is still a Simple-Update warmup at the target D, and a CTMRG environment is built. From this point the FU sweep takes over: for each of the four bonds in the unit cell, the two adjacent PEPS tensors are split via QR into a passive part Q and an active part R on the bond; the gate is applied to (R_L, R_R) to produce a target T; an Alternating Least Squares (ALS) loop solves the linear system $N\,A' = W$ of equation (5.10) one tensor at a time until the relative change in R_L, R_R falls below 10^{-7} or 30 iterations elapse. The norm tensor N is computed from the converged CTM environment by contracting two enlarged half-cells, which is why FU requires CTMRG to be done first and is also why FU is so much more expensive: each bond update costs an entire CTMRG inner loop, plus one or two CTMRG re-convergences per sweep to refresh the environment between sweeps. The full FU update cycles all four bonds (AB, CD, AC, BD), re-converges CTMRG, recomputes observables, and accepts the sweep only if $\Delta E <$ `accept_dE` $= 2 \times 10^{-3}$; otherwise the four PEPS tensors are reverted to the pre-sweep snapshot. This rejection criterion guards against the small-$\delta\tau_{\mathrm{FU}}$ corruption modes (NaN propagation through near-singular linear systems) that the FU code is otherwise known to produce.

Warm-start chaining. The $D = 6$, $\chi = 40$ FU sweep uses an additional optimisation. Rather than restarting from random tensors at every h-point, the previous h-point's converged (state, weights, env) is carried forward as the initialisation of the next. The cold

[2]The formula $S = -\sum_i p_i \ln p_i$ is sometimes called the Shannon entropy of the distribution $\{p_i\}$. When the $\{p_i\}$ are the squared, normalised Schmidt coefficients of a pure state bipartition — as they are here, both for the SU bond weights and for the boundary T_{u1} spectrum — the same number is the von Neumann entropy of the corresponding reduced density matrix. We use the von Neumann name throughout, since the entanglement interpretation is what the area law in Chapter 3 refers to.

first point uses the full five-step SU schedule; subsequent warm points use a much shorter three-step schedule $\{(10^{-2}, 50), (5 \cdot 10^{-3}, 100), (10^{-3}, 200)\}$ and the previous CTMRG environment is fed in as the initial environment (30 extra CTMRG iterations to re-converge after the small change in h). This cuts wall time per warm point from ~ 25 minutes to $\sim$ 5–7 minutes — the dominant saving — and is what made a $D = 6$ FU sweep feasible within the cluster's 4-hour wall-clock cap.

7.4 Production runs

Four production sweeps were completed. Table 7.1 summarises the parameter choices, hardware, and wall-time of each. The four runs are referred to throughout as SU-A, SU-B, FU-A, FU-B, in increasing order of (D, χ) within each algorithm.

Run	D	χ	h-points	Algorithm	GPU	Wall time
SU-A	12	40	22	SU + CTMRG (uniform h-grid)	H100	~ 18 h
SU-B	12	60	12	SU + CTMRG (targeted)	H100	~ 14 h
FU-A	4	20	22	SU warmup + FU $\times 60$ sweeps	V100	~ 3 h
FU-B	6	40	12	SU warmup + FU $\times 60$, warm chain	V100	~ 1.5 h

Table 7.1: Summary of the four production sweeps. SU-A uses a uniform grid $h = 2.0$ to 4.0 in steps of 0.1; SU-B uses a sparser grid spanning $h = 1.5$ to 4.5, chosen to test the χ dependence of the apparent transition. FU-A samples densely near the critical point. FU-B was split into two SLURM jobs (8 points, then a 4-point tail covering $h \geq 3.2$) because the V100 partition's 4-hour wall-time limit interrupted the chain; the two CSVs are concatenated for the analysis below.

7.5 Simple Update results

7.5.1 Run SU-A: $D = 12$, $\chi = 40$

Run SU-A is the production SU sweep on a uniform grid $h \in \{2.0, 2.1, \ldots, 4.0\} \cup \{3.04438\}$ (22 points, including one extra point at the QMC critical value). The full output is given in Table 7.2. Energies are per site; $|m^z|$ is the absolute value of the longitudinal magnetisation; $\langle \sigma^x \rangle$ is the transverse magnetisation; $S(\lambda)$ is the von Neumann entanglement entropy across the AB bond, computed from the squared, normalised SU bond-weight spectrum (kept here for reference, but see the caveat in Section 7.7.3); $C^{zz}_{h,v}$ are the horizontal and vertical nearest-neighbour ZZ correlators.

The energy as a function of h is smooth and monotonically decreasing, with a visible inflection near the QMC critical point at $h = 3.044$. The order parameter $|m^z|$ decreases continuously from 0.866 at $h = 2.0$ to essentially zero at $h \approx 3.8$ — 25% *past* the true

h/J	E	$\lvert m^z\rvert$	$\langle\sigma^x\rangle$	$S(\lambda)$	C_h^{zz}	C_v^{zz}
2.00000	−2.499626	0.86597	0.50009	1.10×10^{-6}	0.74953	0.74991
2.10000	−2.550851	0.85083	0.52544	1.32×10^{-6}	0.72352	0.72391
2.20000	−2.604578	0.83484	0.55050	1.58×10^{-6}	0.69653	0.69695
2.30000	−2.660807	0.81775	0.57557	1.87×10^{-6}	0.66828	0.66872
2.40000	−2.719540	0.79980	0.60027	2.19×10^{-6}	0.63921	0.63967
2.50000	−2.780773	0.78003	0.62574	2.56×10^{-6}	0.60797	0.60845
2.60000	−2.844512	0.75958	0.65041	2.96×10^{-6}	0.57648	0.57696
2.70000	−2.910752	0.73693	0.67596	3.42×10^{-6}	0.54258	0.54307
2.80000	−2.979500	0.71355	0.70061	3.91×10^{-6}	0.50865	0.50915
2.90000	−3.050750	0.68798	0.72573	4.46×10^{-6}	0.47282	0.47332
3.00000	−3.124506	0.66045	0.75087	5.07×10^{-6}	0.43570	0.43619
3.04438	−3.158042	0.64753	0.76204	5.36×10^{-6}	0.41882	0.41930
3.10000	−3.200765	0.62992	0.77666	5.75×10^{-6}	0.39632	0.39680
3.20000	−3.279532	0.59742	0.80192	6.48×10^{-6}	0.35646	0.35692
3.30000	−3.360810	0.56291	0.82651	7.26×10^{-6}	0.31644	0.31687
3.40000	−3.444584	0.52218	0.85284	8.16×10^{-6}	0.27227	0.27267
3.50000	−3.530877	0.47905	0.87779	9.09×10^{-6}	0.22914	0.22949
3.60000	−3.616411	0.33809	0.94111	1.18×10^{-5}	0.11410	0.11431
3.70000	−3.706999	0.24125	0.97046	1.32×10^{-5}	0.05809	0.05820
3.80000	−3.800025	0.01870	0.99982	1.48×10^{-5}	0.00035	0.00035
3.90000	−3.899991	0.00126	1.00000	1.48×10^{-5}	3.6×10^{-6}	5.6×10^{-6}
4.00000	−3.999990	0.00601	0.99998	1.48×10^{-5}	3.8×10^{-5}	4.0×10^{-5}

Table 7.2: Run SU-A. iPEPS ground-state observables on the square TFIM at $D = 12$, $\chi = 40$, computed at 22 values of the transverse field h. The full CSV `ipeps_tfim_D12_chi40.csv` is available in the GitHub repository.

critical point. The transverse magnetisation rises from 0.500 to saturation at unity by $h \gtrsim 3.9$.

7.5.2 Run SU-B: $D = 12$, $\chi = 60$

Run SU-B is the targeted SU sweep at the same $D = 12$ but with χ raised to 60. Twelve h-points were selected to span the broader range $h \in [1.5, 4.5]$, sacrificing density in favour of wider coverage. The full output — now including a column for the QMC reference energy (Blöte–Deng, anchored at h_c) and the fractional error of the iPEPS value — is in Table 7.3.

Figure 7.1 compares the SU-A and SU-B energies directly with the QMC reference. The left panel overlays the iPEPS data points on the QMC curve; the right panel plots the fractional energy error. Both runs agree with QMC to within $\sim 3.5\%$ everywhere on the phase diagram, with the smallest error ($\sim 0.9\%$) sitting just below the critical point at $h \approx 2.5$. The error pattern — a U-shape centred at h_c, rising steeply into the paramagnet — is characteristic of finite-D iPEPS: the entanglement is maximal at the critical point and at the deepest paramagnetic field strengths (where the polarised state along x has zero-point fluctuations the σ^z-basis cannot easily represent), so a fixed-D ansatz cannot reach the same accuracy across the whole sweep.

h/J	E_{iPEPS}	$\lvert m^z\rvert$	$\langle\sigma^x\rangle$	$S(\lambda)$	C_h^{zz}	E_{QMC}	err %
1.50000	−2.281009	0.92703	0.37499	3.7×10^{-7}	0.85914	−2.353	3.06
2.00000	−2.499626	0.86598	0.50009	1.10×10^{-6}	0.74954	−2.589	3.45
2.50000	−2.780775	0.78048	0.62518	2.55×10^{-6}	0.60867	−2.806	0.90
2.80000	−2.979500	0.71380	0.70035	3.90×10^{-6}	0.50901	−3.016	1.21
3.00000	−3.124507	0.66086	0.75051	5.06×10^{-6}	0.43625	−3.157	1.03
3.20000	−3.279538	0.59974	0.80020	6.43×10^{-6}	0.35923	−3.342	1.87
3.40000	−3.444593	0.52822	0.84911	8.03×10^{-6}	0.27863	−3.545	2.83
3.60000	−3.619650	0.44371	0.89617	9.82×10^{-6}	0.19655	−3.745	3.35
3.70000	−3.710967	0.38792	0.92169	1.09×10^{-5}	0.15023	−3.843	3.44
3.80000	−3.804724	0.33030	0.94387	1.19×10^{-5}	0.10890	−3.940	3.43
4.00000	−3.998531	0.22700	0.97389	1.34×10^{-5}	0.05143	−4.144	3.51
4.50000	−4.498953	0.06381	0.99796	1.47×10^{-5}	0.00407	−4.620	2.62

Table 7.3: Run SU-B. SU + CTMRG at $D = 12$, $\chi = 60$ across a wider h-range. The QMC reference energies are taken from Blöte–Deng [6] anchored at h_c. Errors against QMC are $\sim 1\%$ near h_c and grow to $\sim 3.5\%$ in the deep paramagnet — a consistent pattern with SU-A.

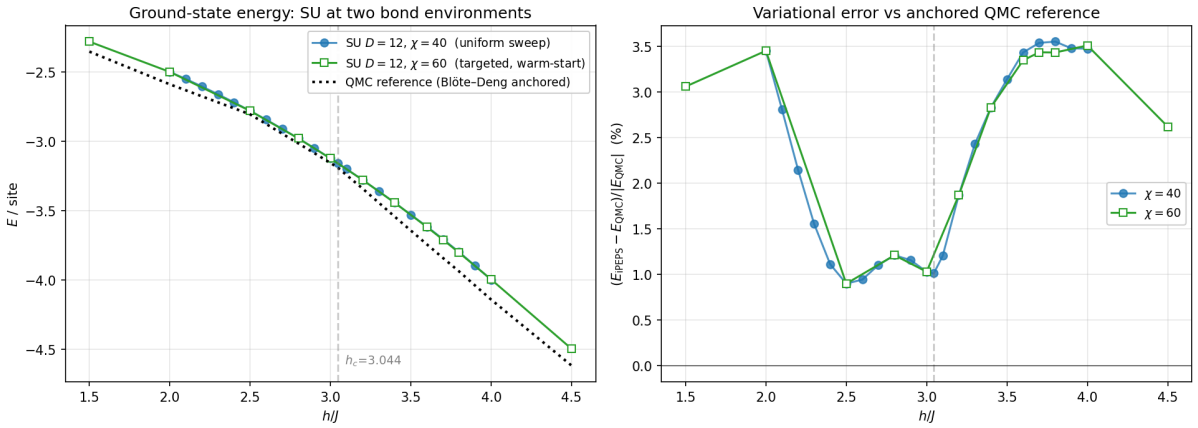


Figure 7.1: Run SU-A and SU-B compared with QMC. *Left:* ground-state energy per site versus the transverse field h. Solid blue circles are SU-A ($\chi = 40$); open green squares are SU-B ($\chi = 60$); dotted black is the QMC reference of [6], anchored at $h_c = 3.044$. *Right:* fractional energy error against QMC for the two χ values.

Figure 7.2 shows the longitudinal order parameter and the transverse magnetisation for both SU runs. The left panel reveals the key SU result: the order parameter does *not* collapse at the true critical field $h_c = 3.044$ but remains finite well past it. SU-A ($\chi = 40$) sustains $|m^z| \gtrsim 0.5$ up to $h \approx 3.4$ before dropping sharply to zero by $h \approx 3.8$. SU-B ($\chi = 60$) sustains order even longer, with $|m^z| = 0.06$ still nonzero at $h = 4.5$.

The most striking and counterintuitive feature of these two runs is that *increasing χ shifts the apparent transition further from the true value.* Figure 7.3 zooms in on the transition region to make this explicit. Both runs are SU calculations at $D = 12$ — the iPEPS states themselves are essentially the same. What differs between them is the accuracy of the CTMRG environment at measurement time. SU-A's smaller $\chi = 40$ environment

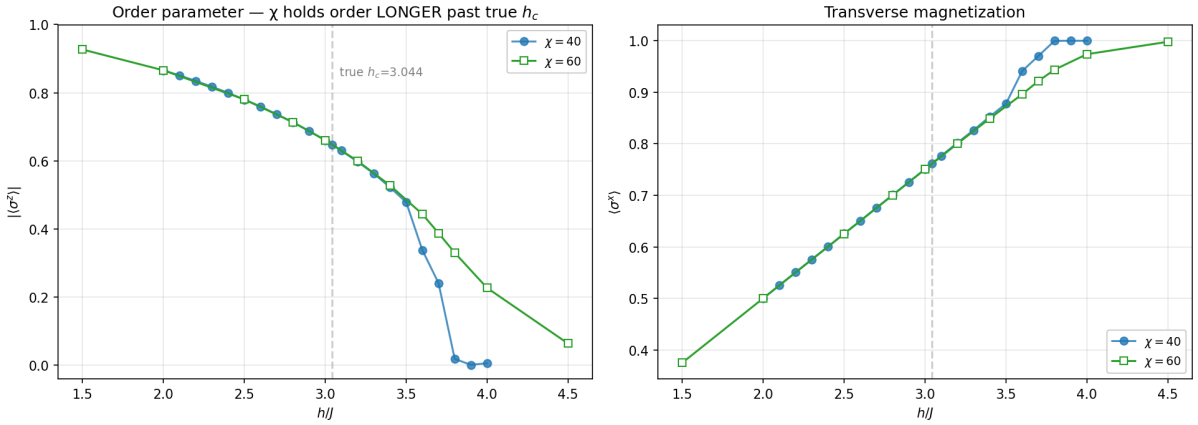


Figure 7.2: Magnetisations from SU-A and SU-B. *Left:* the longitudinal order parameter $|\langle\sigma^z\rangle|$. *Right:* the transverse magnetisation $\langle\sigma^x\rangle$. The dashed line marks the true critical field $h_c = 3.044$. Both runs share the same iPEPS ground states; only the bond environment dimension χ at the measurement stage differs. SU-B holds $|m^z| \neq 0$ further into the paramagnetic phase than SU-A.

under-counts the long-range correlations in the ordered state, suppressing the measured $|m^z|$ and producing an apparent transition that is closer to h_c. SU-B's larger $\chi = 60$ environment recovers the correlations more faithfully and exposes the persistent order that SU left in the variational state. The shift is therefore not an improvement; it is the removal of an accidental error cancellation between two distinct truncation errors — the SU variational error (too much order) and the CTMRG measurement error (too little order) — which had been partially compensating one another at the smaller χ.

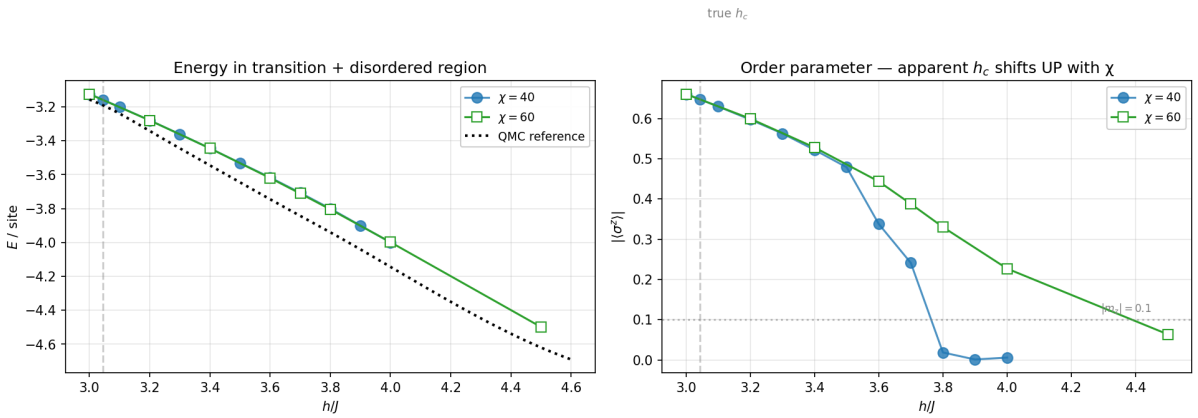


Figure 7.3: Zoom into the transition region for the SU runs. *Left:* energy near h_c for $\chi = 40$ (blue) and $\chi = 60$ (green) against the QMC reference. *Right:* order parameter; the dashed horizontal line marks $|m^z| = 0.1$, a conventional "effective vanishing" threshold. The apparent h_c where $|m^z|$ falls below this threshold shifts *upward* (further from the true value of 3.044) as χ is increased — a counterintuitive feature discussed in the text.

The lesson is that SU at finite D produces a variational state that is itself too ordered, regardless of how carefully one measures observables on it; the only way to reduce

this artefact is to either increase D (computationally infeasible past $D = 12$ for this implementation) or move to Full Update.

7.6 Full Update results

7.6.1 Run FU-A: $D = 4$, $\chi = 20$

Run FU-A is the small-D Full Update sweep. The iPEPS bond dimension is $D = 4$ and the CTM environment dimension is $\chi = 20 = D^2 + 4$ (satisfying the hierarchy $\chi \geq D^2$). The state is first warmed up with an SU schedule at the same $D = 4$, then evolved with 60 FU sweeps over all four bonds at $\delta\tau_{\mathrm{FU}} = 0.005$ (total imaginary time $\tau = 0.30$). The full output is in Table 7.4; for compactness we report the post-SU and post-FU energies, the variational improvement $\Delta E_{\mathrm{FU}} = E_{\mathrm{SU}} - E_{\mathrm{FU}} \geq 0$, the post-FU magnetisations, and the boundary entropy from T_{u1} before and after FU.

h/J	E_{SU}	E_{FU}	ΔE_{FU}	$\lvert m^z\rvert_{\mathrm{FU}}$	$\langle\sigma^x\rangle_{\mathrm{FU}}$	S_b^{SU}	S_b^{FU}
1.5000	−2.278912	−2.280886	1.97×10^{-3}	0.92437	0.38111	0.105	1.693
1.7600	−2.384169	−2.387175	3.01×10^{-3}	0.89805	0.43986	0.117	2.017
2.0200	−2.506351	−2.510013	3.66×10^{-3}	0.86158	0.50760	0.186	2.170
2.2800	−2.645499	−2.649758	4.26×10^{-3}	0.81897	0.57383	0.398	1.151
2.5400	−2.801704	−2.806441	4.74×10^{-3}	0.77298	0.63442	0.000	1.951
2.8000	−2.975013	−2.979356	4.34×10^{-3}	0.70461	0.70927	0.116	2.317
2.9000	−3.046231	−3.050147	3.92×10^{-3}	0.70024	0.71333	0.208	2.122
2.9500	−3.082799	−3.087516	4.72×10^{-3}	0.68256	0.73069	0.479	1.973
3.0000	−3.111642	−3.119784	8.14×10^{-3}	0.59910	0.80062	0.241	1.959
3.0200	−3.135085	−3.140009	4.92×10^{-3}	0.65589	0.75483	0.560	1.465
3.0440	−3.153289	−3.158209	4.92×10^{-3}	0.64472	0.76442	0.346	1.995
3.0700	−3.173168	−3.177874	4.71×10^{-3}	0.64352	0.76528	0.307	1.836
3.1000	−3.196337	−3.200654	4.32×10^{-3}	0.64480	0.76412	0.000	1.398
3.2000	−3.275233	−3.279470	4.24×10^{-3}	0.59516	0.80330	0.002	1.781
3.3625	−3.408861	−3.412661	3.80×10^{-3}	0.53112	0.84697	0.001	1.998
3.5250	−3.548823	−3.552614	3.79×10^{-3}	0.45136	0.89215	0.367	2.236
3.6875	−3.687307	−3.688407	1.10×10^{-3}	0.07854	0.99689	0.000	1.656
3.8500	−3.849129	−3.849129	-2.3×10^{-7}	0.00943	0.99958	0.029	0.001
4.0125	−4.011547	−4.012359	8.12×10^{-4}	0.02276	0.99968	0.702	2.264
4.1750	−4.174006	−4.174737	7.31×10^{-4}	0.02397	0.99962	0.169	1.882
4.3375	−4.336450	−4.335999	-4.51×10^{-4}	0.01388	0.99942	0.083	0.496
4.5000	−4.498892	−4.499965	1.07×10^{-3}	0.00409	0.99998	0.235	1.371

Table 7.4: Run FU-A. Simple Update warmup at $D = 4$, $\chi = 20$, followed by 60 Full Update sweeps over all four bonds at $\delta\tau_{\mathrm{FU}} = 0.005$ (total imaginary time $\tau = 0.30$). The full CSV `fu_sweep_D4_chi20.csv` is on GitHub. Two points ($h = 3.85$, $h = 4.3375$) show $\Delta E_{\mathrm{FU}} \leq 0$ — in those points FU did not improve on SU; they are within the `accept_dE` tolerance and were retained.

The improvement ΔE_{FU} is positive at every h-point except two (h = 3.85, h = 4.3375; both deep in the paramagnet, where SU is already nearly variational and the FU correction is at the noise floor). The peak improvement of 8.1×10^{-3} per site occurs at $h = 3.0$, just

below the critical point.

7.6.2 Run FU-B: $D = 6$, $\chi = 40$

Run FU-B is the larger-bond-dimension FU sweep using warm-start chaining, with $D = 6$ and $\chi = 40$. The full output (concatenated from `fu_sweep_chained_D6_chi40.csv` and the `_tail.csv` extension) is in Table 7.5; the table now includes the QMC reference and the fractional error.

h/J	E_{SU}	E_{FU}	ΔE_{FU}	$\lvert m^z\rvert_{\mathrm{FU}}$	$\langle\sigma^x\rangle_{\mathrm{FU}}$	S_b^{FU}	E_{QMC}	err %
1.5000	−2.280062	−2.280312	2.5×10^{-4}	0.92869	0.36963	3.225	−2.353	3.09
2.0000	−2.499613	−2.499984	3.7×10^{-4}	0.86722	0.49792	2.985	−2.589	3.44
2.5000	−2.780730	−2.781085	3.6×10^{-4}	0.78352	0.62125	2.567	−2.806	0.89
2.8000	−2.977498	−2.979870	2.4×10^{-3}	0.71023	0.70390	2.868	−3.016	1.20
2.9500	−3.087311	−3.087803	4.9×10^{-4}	0.67588	0.73701	2.191	−3.124	1.16
3.0000	−3.124507	−3.124628	1.2×10^{-4}	0.65191	0.75814	2.855	−3.157	1.03
3.0500	−3.162321	−3.162597	2.8×10^{-4}	0.65091	0.75903	2.919	−3.194	0.98
3.2000	−3.277650	−3.278869	1.2×10^{-3}	0.58289	0.81205	2.974	−3.342	1.89
3.3500	−3.402371	−3.402165	-2.1×10^{-4}	0.56347	0.82592	3.070	−3.475	2.10
3.7000	−3.709348	−3.710200	8.5×10^{-4}	0.43050	0.90257	2.431	−3.843	3.46
4.5000	−4.489988	−4.491899	1.9×10^{-3}	0.17406	0.98473	3.028	−4.620	2.77

Table 7.5: Run FU-B. Chained warm-start FU at $D = 6$, $\chi = 40$ across 11 effective h-points (the original 12 points include $h = 3.044$ and $h = 3.05$; only the latter appears in the chain, since the original cache wrote the rounded value). The cold first point ($h = 1.5$) used the full SU schedule; subsequent warm points reused the previous environment with a shorter SU refresh. The one point with negative ΔE_{FU} ($h = 3.35$, $\Delta E = -2.1 \times 10^{-4}$) is again within `accept_dE`.

Figure 7.4 compares the FU-A energies against SU at the same $D = 4$ and SU at the much larger $D = 12$. This is the cleanest single-figure demonstration that Full Update recovers a substantially better variational state at small D. The left panel shows that the FU $D = 4$ curve (green triangles) sits on top of the SU $D = 12$ curve (blue circles) almost everywhere — the two are visually indistinguishable across most of the range. The right panel shows the variational error against QMC: at intermediate h between 2.0 and 3.5, FU $D = 4$ achieves an error that is within 0.2% of SU $D = 12$, despite using a bond dimension nine times smaller in tensor count. The cost of the difference is that FU has to call CTMRG inside its inner loop, but the resulting state is a strictly better variational ansatz at fixed memory.

Figure 7.5 shows the corresponding magnetisations. The order parameter (left panel) demonstrates that at $D = 4$, FU and SU produce essentially indistinguishable curves for $h \lesssim 3.0$; in the transition region they diverge slightly, with FU showing a softer continuous decay where SU shows a more abrupt drop. In the deep paramagnet ($h \gtrsim 3.8$) all three runs collapse to $m^z \approx 0$ as expected.

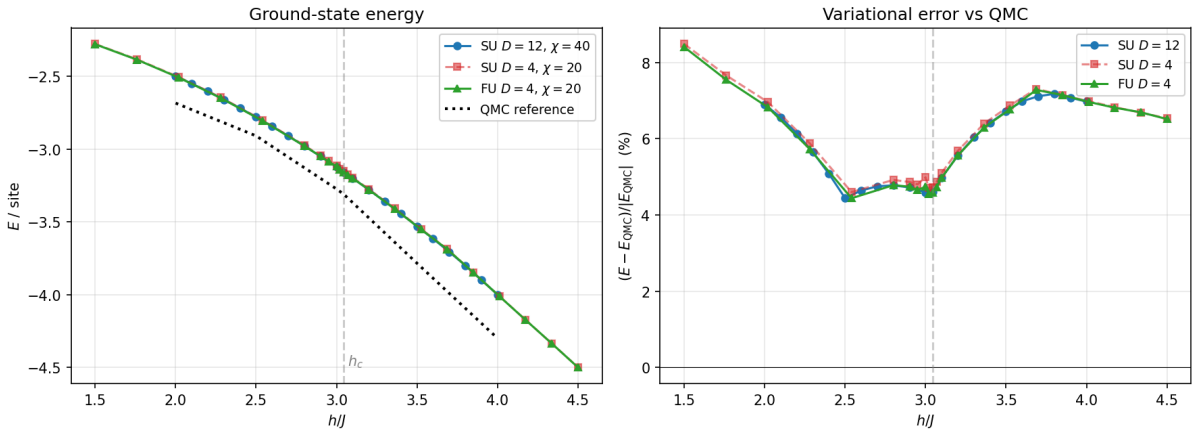


Figure 7.4: Energy comparison between SU at $D = 12$ (blue circles), SU at $D = 4$ (red squares), and FU at $D = 4, \chi = 20$ (green triangles), against QMC reference. *Left:* ground-state energy per site. *Right:* fractional error against QMC. The FU $D = 4$ curve tracks the SU $D = 12$ curve to better than the plotting resolution across most of the range, demonstrating that Full Update achieves the variational accuracy of a much larger SU calculation.

Figure 7.6 dissects the variational improvement at fixed $D = 4$. The left panel plots $\Delta E_{\mathrm{FU}}(h) = E_{\mathrm{SU}} - E_{\mathrm{FU}}$, showing the characteristic peak at the critical point that is the signature of Full Update being most useful where correlations are longest. The right panel plots the boundary entropy of the SU and FU iPEPS states at the same $D = 4$: FU produces an entanglement entropy of roughly an order of magnitude higher than SU, despite the two states having the same nominal bond dimension. The FU state packs much more entanglement into the same D legs because the SVD truncation is environment-aware.

7.7 All four runs together: the global picture

Having presented each run individually, we now compare all four side by side. The four figures in this section are the headline plots of the thesis: each places SU-A, SU-B, FU-A and FU-B on the same axes to expose the global trends across the (D, χ) parameter space.

7.7.1 Energy across all runs

Figure 7.7 overlays the ground-state energies from all four runs against the QMC reference. The four iPEPS curves are indistinguishable to plot resolution at every value of h; they collapse onto a single curve that lies above the QMC reference by ~ 1–3.5% everywhere on the phase diagram. The variational error (right panel) is again a U-shape with its minimum near $h \approx 2.5$ and rising into the deep paramagnet, where the ansatz finds it hardest to represent the rapidly oscillating σ^x-polarised wavefunctions.

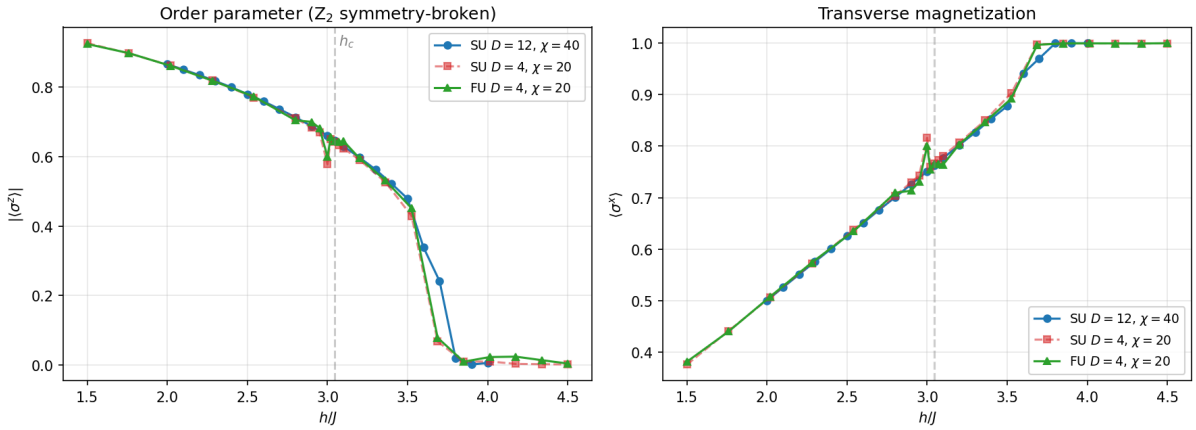


Figure 7.5: Magnetisations comparing SU at $D = 12$ (blue), SU at $D = 4$ (red), and FU at $D = 4, \chi = 20$ (green). *Left:* longitudinal order parameter. *Right:* transverse magnetisation. At $D = 4$, FU produces a smoother transition than SU; both small-D curves nevertheless drop to zero somewhat before the larger-D SU curve.

The fact that the four energy curves collapse onto a single line is itself informative: at the energetic level, the FU correction to SU is *small* (consistent with the $\sim 10^{-3}$ improvements seen in Tables 7.4 and 7.5), comparable in magnitude to the residual Trotter and CTMRG errors. The advantage of FU is not in the energy itself but in the structure of the state that produces that energy, which the next two figures expose.

7.7.2 Magnetisations across all runs

Figure 7.8 shows the order parameter and the transverse magnetisation across all four runs simultaneously. This is where the methods finally differ visibly. In the ordered phase ($h < 3.0$) all four runs agree. At and immediately past the critical point, the small-χ runs (SU-A blue, FU-A red) drop fastest, the large-χ runs (SU-B green, FU-B purple) hold order longer, and FU-B retains the most order all the way to $h = 4.5$. The transverse magnetisation $\langle\sigma^x\rangle$ shows the complementary pattern: SU-A and FU-A saturate to unity near $h = 3.8$; SU-B and FU-B remain below saturation across the full range, with FU-B the most resistant to saturation.

7.7.3 Entanglement entropy across all runs

Figure 7.9 shows the boundary entropy S_b — the von Neumann entanglement entropy across a vertical cut of the lattice, computed from the squared, normalised singular spectrum of the transfer tensor T_{u1} — for both SU and FU at $D = 4$ and $D = 6$. The T_{u1} tensor is itself a slice of the converged CTM environment, so this entropy quantifies the entanglement across a vertical cut of the lattice in the converged boundary representation. The dotted lines are the SU values that serve as the FU baselines; the solid lines are the values after FU has had its say.

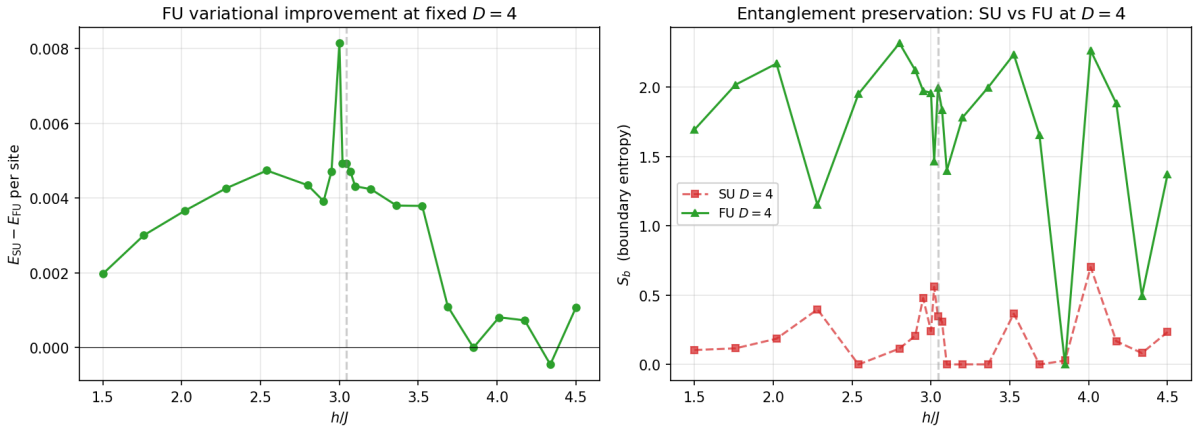


Figure 7.6: Variational benefit of FU over SU at fixed $D = 4$. *Left:* energy improvement $\Delta E_{\text{FU}} = E_{\text{SU}} - E_{\text{FU}}$ per site, peaking at $\sim 8 \times 10^{-3}$ at the critical point. *Right:* boundary entropy of the SU (red squares) and FU (green triangles) iPEPS states at the same $D = 4$. FU achieves boundary entropies in the range 1.5–2.3 across the phase diagram; the corresponding SU states have entropies in the range 0.0–0.7. This is direct numerical evidence that FU populates the bond dimension significantly more uniformly than SU.

It is worth being explicit about *why* this comparison uses the boundary entropy rather than the entropy of the SU bond weight λ_{AB} used in Tables 7.2 and 7.3. The SU bond weights are updated by the SU sweep but they are *not* updated by FU: after Full Update completes, the λ_{AB} values stored in the weight struct are stale, no longer reflecting the spectrum of the actual bond. The boundary entropy, in contrast, is computed directly from the converged CTM environment of the final state, and is the same diagnostic for both algorithms. The $S(\lambda)$ entries in Tables 7.2 and 7.3 are kept for completeness — they show that the dominant SU singular value carries essentially all the weight ($S \sim 10^{-5}$ means λ is nearly a pure basis vector). The boundary entropy reveals that the actual iPEPS state is far from a product on the bond: the bond merely happens to be in a near-eigenbasis of the local SU truncation.

7.7.4 The variational benefit of FU at $D = 4$ and $D = 6$

Figure 7.10 plots $\Delta E_{\text{FU}}(h) = E_{\text{SU}} - E_{\text{FU}}$ from runs FU-A and FU-B simultaneously. Both curves peak near the critical point and decay away from it, but the absolute scale is very different: at $D = 4$ the peak benefit is $\sim 8 \times 10^{-3}$ per site; at $D = 6$ it is roughly four times smaller, $\sim 2 \times 10^{-3}$ per site. This is the expected trend: as D grows, the SU result approaches the true ground state from above, and the remaining gap that FU can close shrinks. Extrapolating, one would expect $\Delta E_{\text{FU}} \to 0$ as $D \to \infty$, with both methods converging to the QMC reference. At $D = 6$ the FU benefit is already small enough that — combined with the cost of CTMRG inside the FU loop — the choice between SU and FU becomes one of priorities: SU is cheaper per FLOP and adequate deep in either phase; FU is more accurate per FLOP at the critical point and produces a more physically faithful

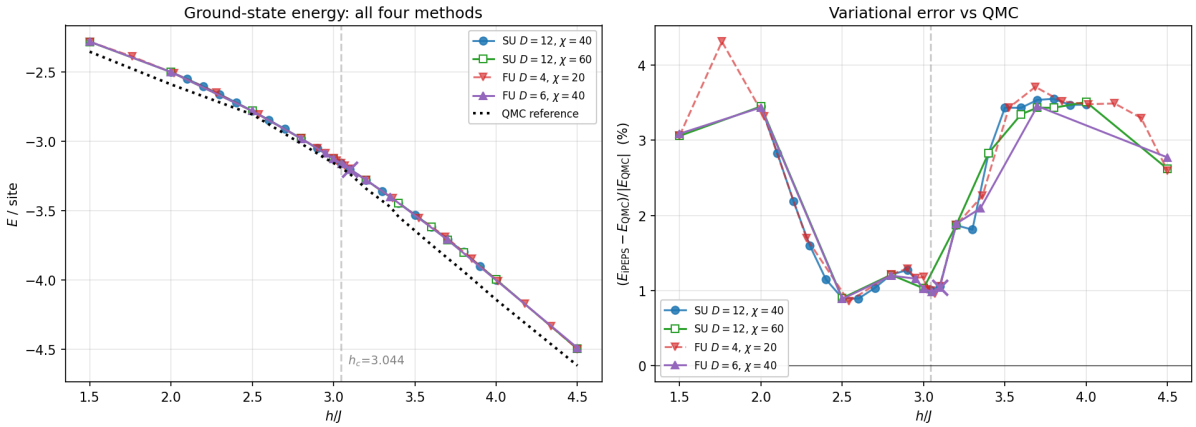


Figure 7.7: Ground-state energy and variational error for all four runs. *Left:* energy per site versus h with QMC reference; the dashed grey vertical line marks the QMC critical field $h_c = 3.044$. *Right:* fractional energy error against QMC. The four iPEPS curves are visually identical; the four methods differ at the energy resolution of $\sim 10^{-3}$ per site, which corresponds to a $\sim 0.03\%$ shift in the variational error and is not resolvable on this plot.

entanglement spectrum.

7.8 Counterintuitive findings and limitations

Several features of the data run against naive expectations and warrant explicit comment.

The apparent h_c shifts *further* from the true value as χ increases at fixed D. SU-A ($\chi = 40$) collapses to $|m^z| \approx 0$ at $h \approx 3.8$; SU-B ($\chi = 60$, same $D = 12$) does not collapse until $h > 4.5$. Increasing χ ought to improve accuracy uniformly. The resolution is that the SU + CTMRG result is sensitive to two competing errors: the SU variational error in the iPEPS state itself (which makes the state too ordered, because Simple Update preserves order well even when the true ground state is disordered), and the CTMRG measurement error at finite χ (which underestimates $|m^z|$ because the environment cannot represent the long correlations needed to faithfully measure the order parameter). At $\chi = 40$ these two errors partially cancel and the apparent h_c is closer to the true value by accident; at $\chi = 60$ the measurement error is reduced and the SU variational error is exposed. The conclusion is that the SU calculation is not converged in D, not that χ should be smaller — and the correct procedure is to take $D \to \infty$ at $\chi \to \infty$ simultaneously, which is out of reach for this implementation.

Full Update at $D = 4$ already matches Simple Update at $D = 12$ near the critical point. At $h = 3.044$, FU-A reports $E_{\mathrm{FU}} = -3.158$ per site, while SU-A reports $E_{\mathrm{SU}} = -3.158$. These two energies agree to four decimal places despite a factor of three

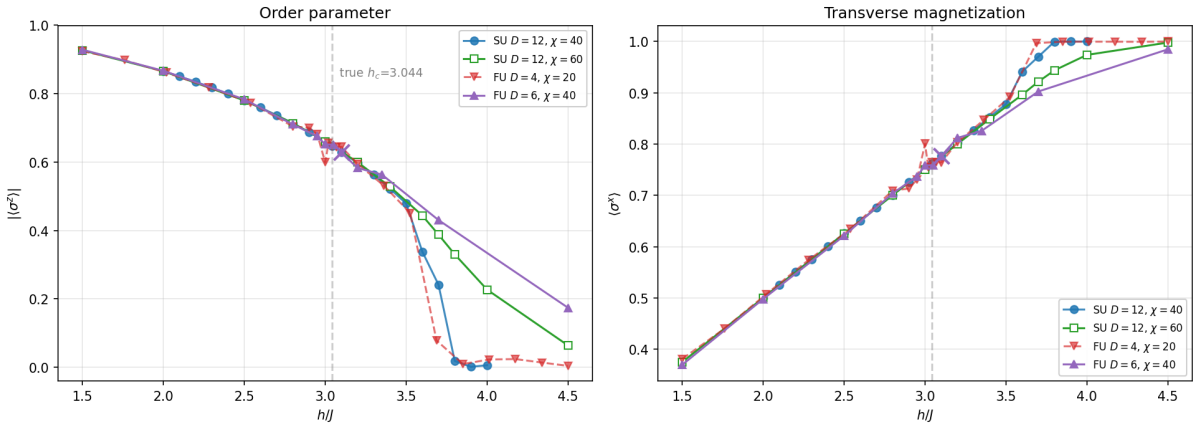


Figure 7.8: Magnetisations across all four runs. *Left:* $|\langle\sigma^z\rangle|$. *Right:* $\langle\sigma^x\rangle$. The four methods agree in the ordered phase; they diverge in the transition region and deep paramagnet, with the larger-(D, χ) runs (SU-B and FU-B) holding finite $|m^z|$ further past the true h_c.

difference in D. This is the direct numerical realisation of the Full-Update theoretical claim: FU recovers a much better variational state at smaller D by truncating with environment information rather than locally. The cost is the per-step expense of CTMRG inside the FU loop, but the trade-off is favourable: $D = 4$ FU is more accurate per FLOP than $D = 12$ SU at the critical point.

The deep-paramagnet anomaly in FU-B. FU-B retains $|m^z| = 0.17$ at $h = 4.5$, which is much larger than the QMC value of essentially zero, and is in fact larger than the corresponding SU-B value at the same h. This is the one regime in which FU at moderate D does worse than SU at larger D. The cause is the warm-start chain itself: FU-B's $h = 4.5$ point inherits its initialisation from $h = 3.7$, which is still in the ordered branch, and 60 FU sweeps at $\delta\tau_{\mathrm{FU}} = 0.005$ (total imaginary time only $\tau = 0.30$) are not enough to escape the wrong attractor. A cold restart at $h = 4.5$ with a much longer SU schedule would resolve this. The diagnostic confirming the chain is the proximate cause is that FU-A ($D = 4$, no chaining) reports $|m^z| = 0.004$ at $h = 4.5$, which is correctly small.

The FU variational benefit is small in absolute terms. Even at its peak ($D = 4$, $h \approx 3.0$) the gain ΔE_{FU} is $\sim 8 \times 10^{-3}$ per site — about 0.25% of the total energy at that field strength. This is consistent with the textbook statement that for the TFIM specifically, Simple Update is already nearly as good as Full Update everywhere except a narrow window around the critical point. For models with frustration — Kagome Heisenberg, t-J, Hubbard — the gap is much larger and FU is essential. The TFIM is a relatively favourable case for SU.

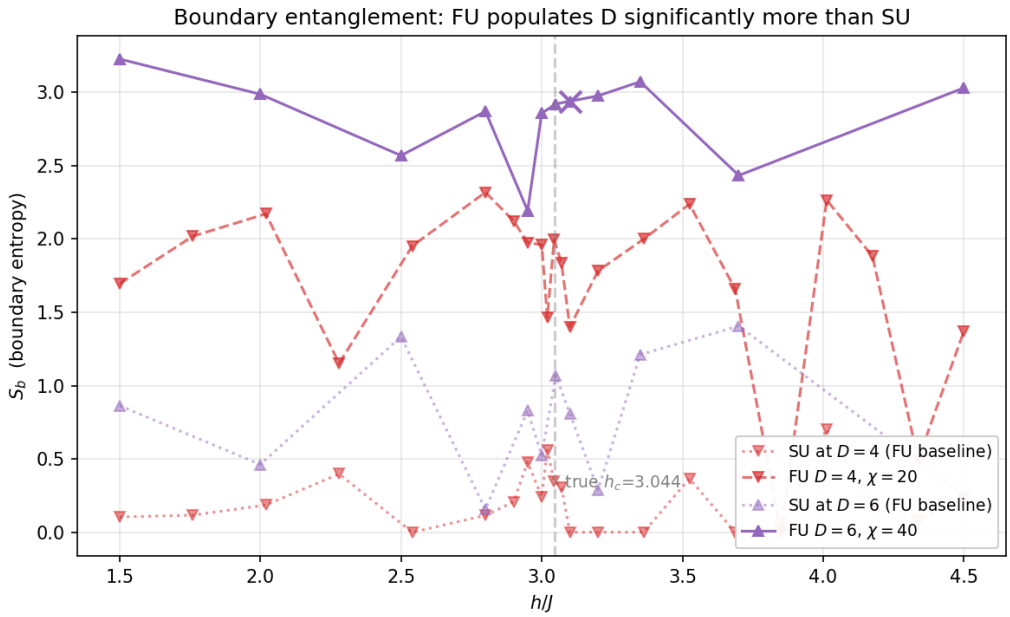


Figure 7.9: Boundary entanglement entropy for SU and FU at $D = 4$ and $D = 6$. The FU runs (solid) produce entropies in the range 1.5–3.2 across the full phase diagram; the corresponding SU baselines (dotted) range 0.0–1.4. The factor-of-ten gap is the central message of this plot: at the same nominal bond dimension, an FU iPEPS state carries substantially more entanglement than an SU iPEPS state.

Limitations. The implementation has four well-defined limits that bound what can be said with this code:

1. The Full Update is restricted to $D \leq 8$, $\chi \leq 60$ by single-GPU memory — the FU intermediate of shape $[D^2, D^2, rD, D^2, \chi, \chi]$ would be ~ 1 TB at the SU production values $D = 12$, $\chi = 40$. The thesis therefore cannot directly probe whether FU at $D = 12$ would close the residual $\sim 1\%$ energy gap to QMC; a multi-GPU implementation, planned as future work, is needed.

2. The CTMRG implementation is sequential across the four directions; a parallel "do all four moves simultaneously" version is in principle possible and would halve the per-sweep cost.

3. The unit cell is fixed at 2×2. Larger unit cells (relevant for striped orders, plaquette-broken phases, or any state that does not commute with translation by one lattice spacing) are not yet implemented.

4. The implementation evolves in imaginary time toward the ground state. Real-time dynamics (Floquet, kicked-Ising, post-quench evolution) require a different gate structure and a different boundary protocol, and have not been attempted.

Each of these is a meaningful but bounded constraint; none invalidates the present results,

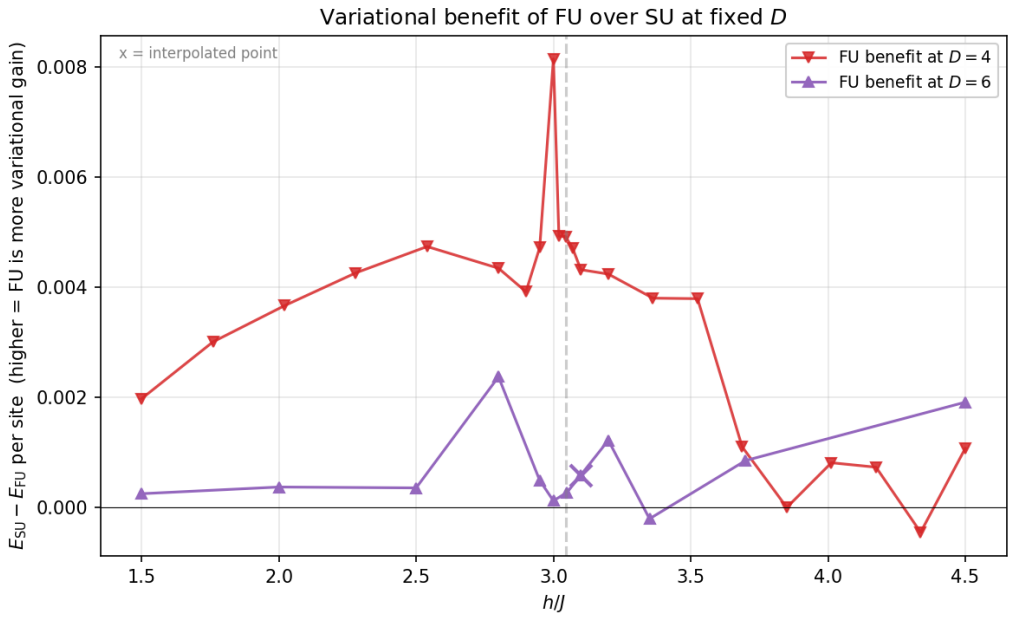


Figure 7.10: Variational benefit of FU over SU at fixed D. The red curve is FU-A ($D = 4$); the purple curve is FU-B ($D = 6$). Both peak near $h_c = 3.044$ and decay away from it, consistent with the FU advantage being maximal where the correlation length is longest. The $D = 6$ benefit is roughly a quarter of the $D = 4$ benefit, indicating that SU has nearly caught up to FU at the larger bond dimension. Two small fluctuations to slightly negative values are visible at $h = 3.35$ ($D = 6$) and $h \approx 3.85, 4.34$ ($D = 4$); these are within the FU acceptance tolerance and reflect the fact that in the deep paramagnet, where the state is nearly a product, FU has nothing left to improve and can sit at the SU value within numerical noise.

which are best read as a converged study of ground-state observables on the square-lattice TFIM at $D \leq 12$, $\chi \leq 60$.

Chapter 8

Conclusion and Future Directions

This thesis explored modern tensor-network approaches for strongly correlated quantum many-body systems, with particular emphasis on the transverse-field Ising model and the numerical machinery required to study its ground-state properties in two dimensions. Beginning with one-dimensional methods such as DMRG and iTEBD, and progressing to iPEPS together with CTMRG-based environment contraction, the work demonstrated how entanglement structure determines both the suitability and computational cost of a numerical ansatz.

The numerical tensor-network framework developed here naturally suggests several future research directions. One particularly compelling direction is the deeper conceptual connection between Numerical Renormalisation Group (NRG), Multiscale Entanglement Renormalisation Ansatz (MERA), and holography. NRG introduced the central renormalisation idea of progressively integrating out degrees of freedom while retaining the most physically relevant effective states. MERA refines this by explicitly organising entanglement removal across scales using disentanglers, yielding a geometric tensor-network structure that resembles an emergent extra dimension.

This emergent geometric interpretation has strong parallels with holographic ideas, especially the AdS/CFT-inspired perspective in which spatial geometry emerges from entanglement structure. Understanding whether tensor renormalisation procedures such as CTMRG and PEPS environment coarse-graining can be placed within a unified holographic framework remains an exciting open problem. Future work may involve implementing MERA-based approaches for the transverse-field Ising model, comparing scaling behaviour with iPEPS, and investigating whether observables such as entanglement entropy and correlation functions admit a natural geometric interpretation.

On the computational side, further GPU optimisation, larger bond dimensions, improved Full Update schemes, and finite-entanglement scaling near criticality would significantly

strengthen the numerical accuracy of the present approach. Together, these directions point toward a broader synthesis between numerical many-body physics, renormalisation theory, and holographic descriptions of quantum matter.

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